Chapter 8：Pattern transfer-etching

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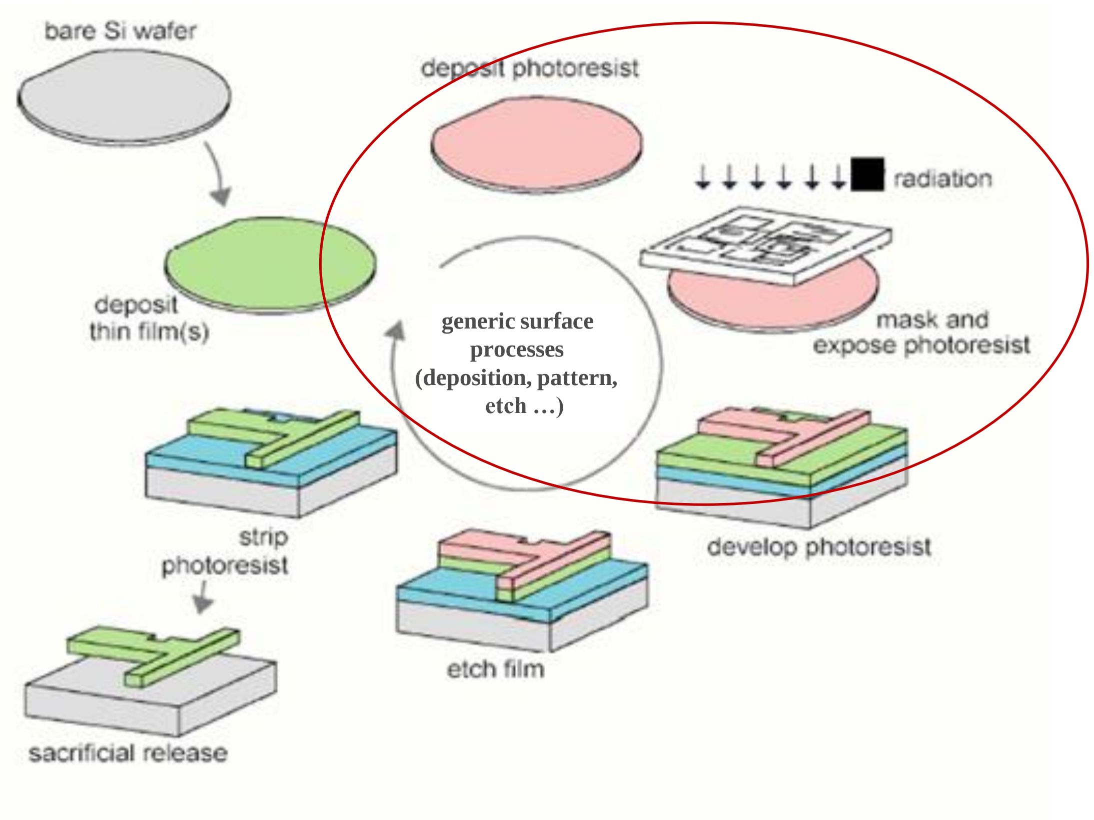
Outline

Planar Processing



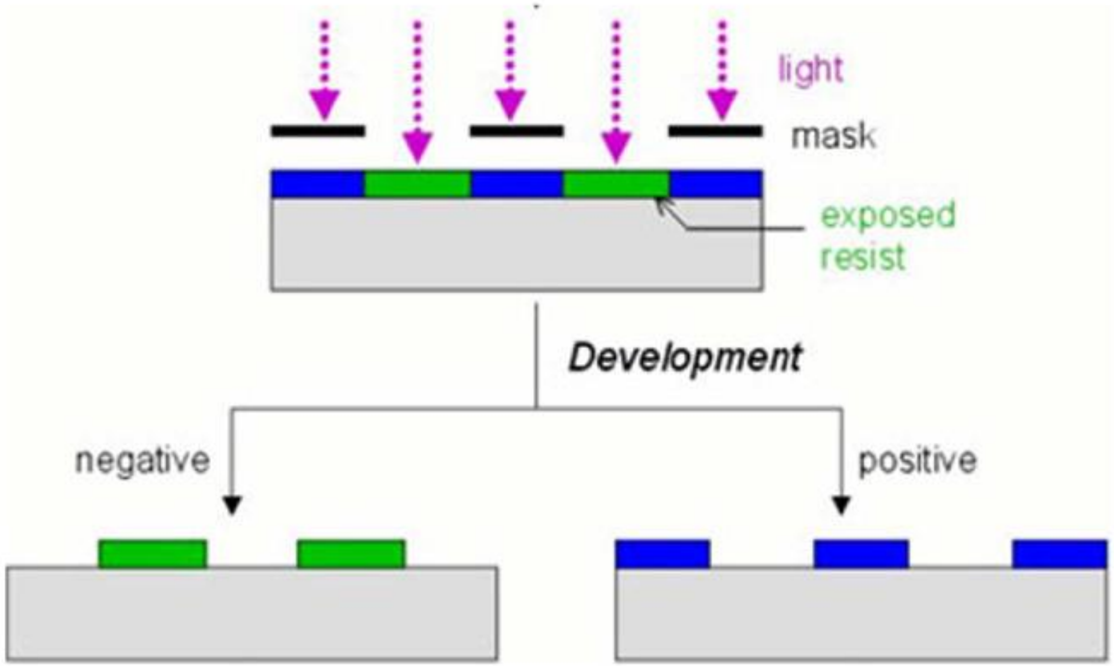
图形转移=光刻+刻蚀

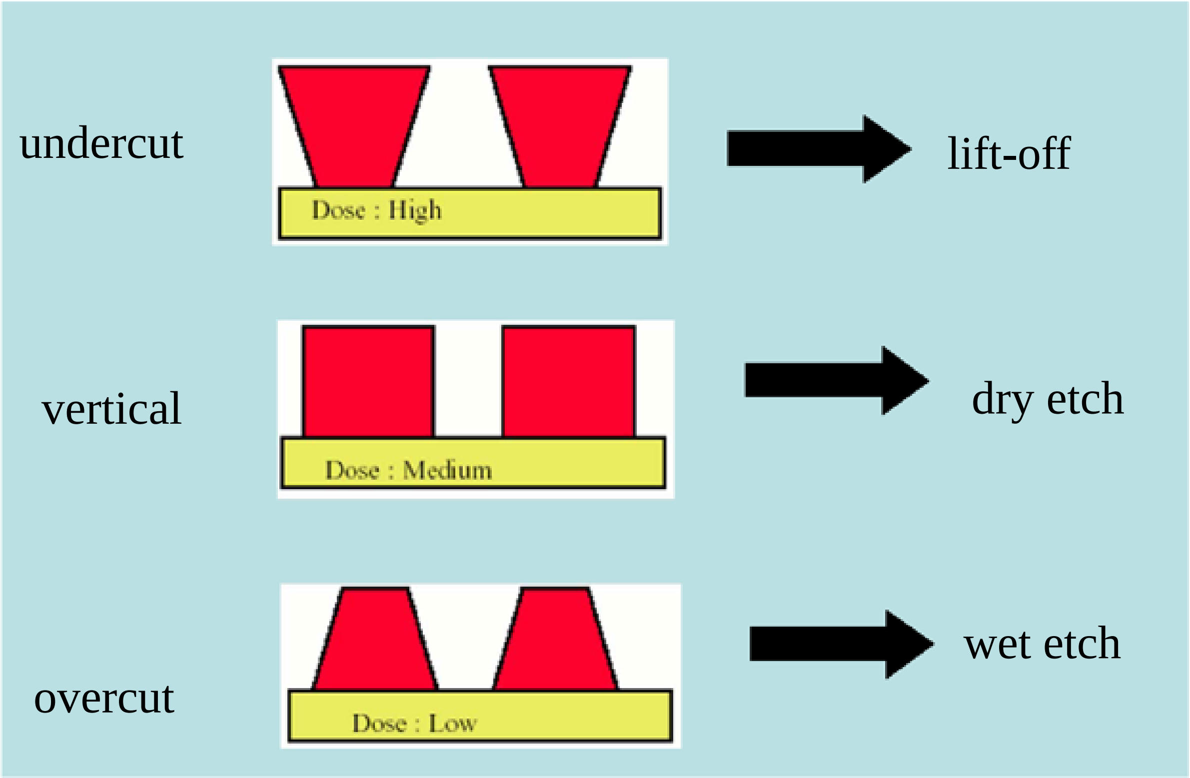
Patterning formation-lithography

Resist Profiles

•Depends on exposure and development

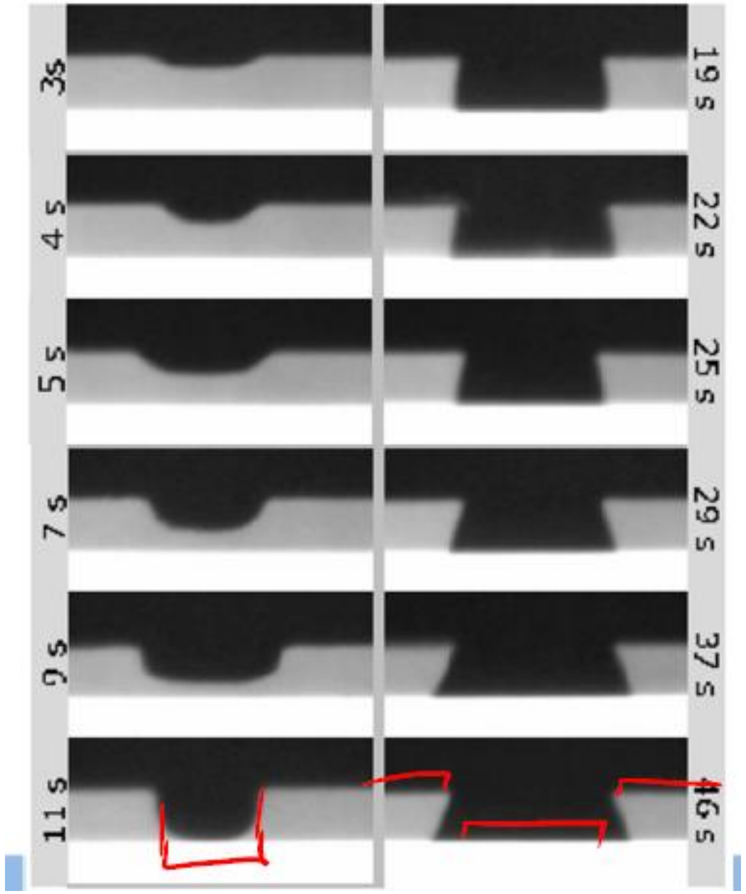
•Can facilitate (or hinder) subsequent processing





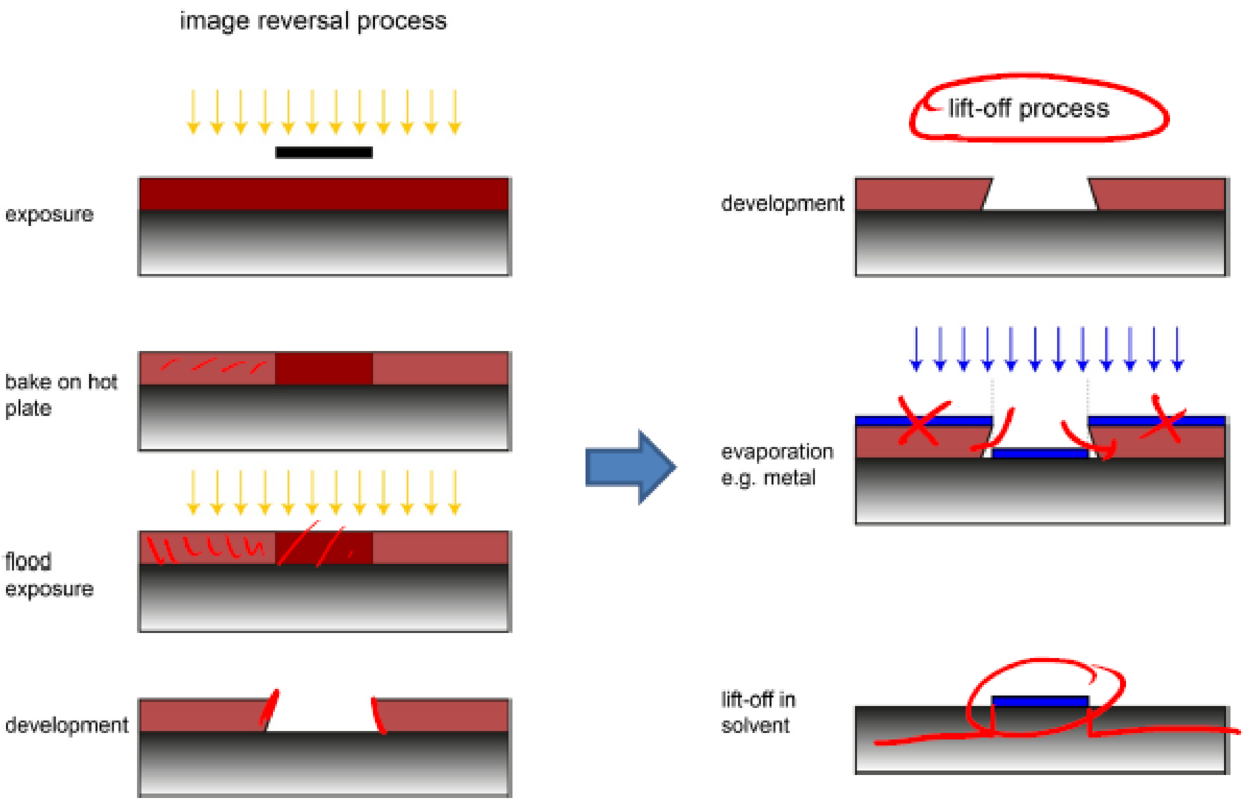
Lithography process

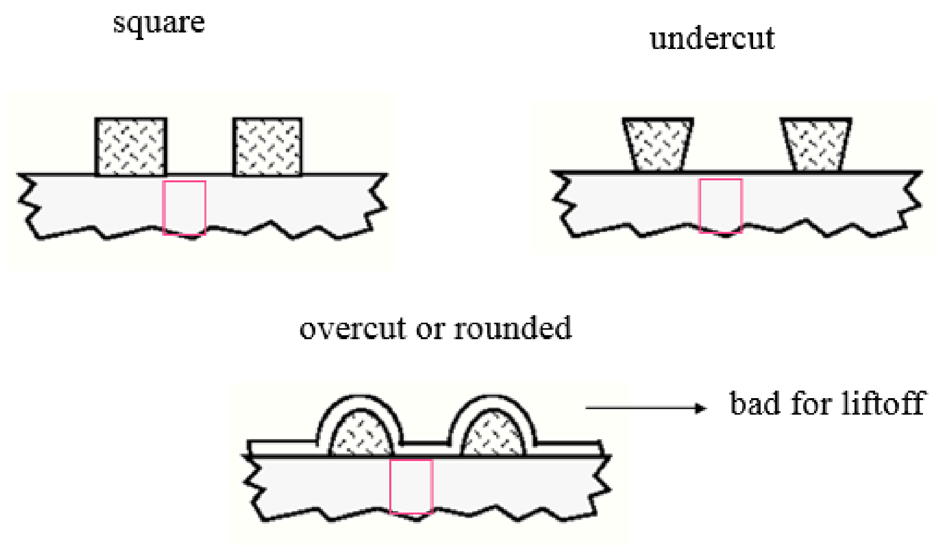
development during positive process



development during image reversal process

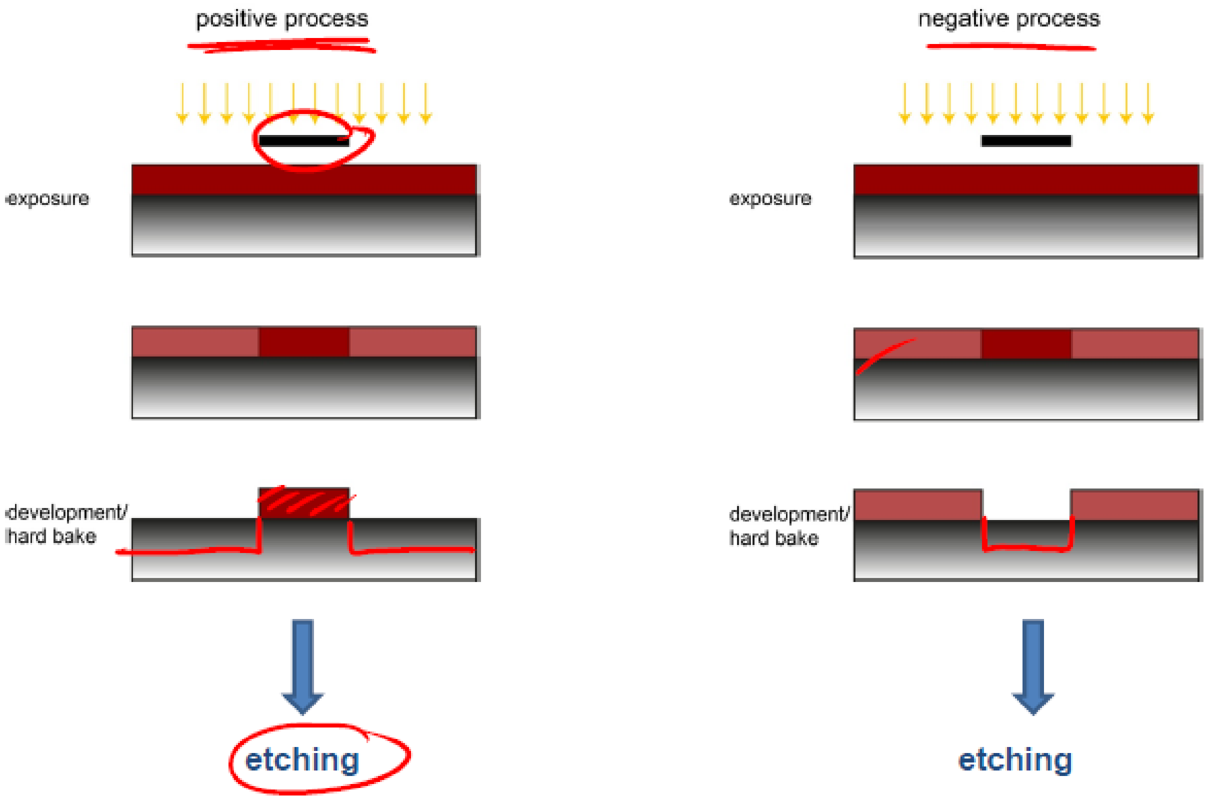
Patterning process-lift off





Resist solvent:
acetone
MEK
methylene chloride
NMP
etc.

Patterning process-etching



Liftoff vs. Etching

- Dependencies:
  - Process flow
  - Materials
- Liftoff:
  - More gentle (in general)
  - Wider range of materials
    - but easier with metals
- Etching:
  - More applicable to thicker films

Advantages of liftoff:

•High resolution

•Limited only by lithography

•Amenable to materials that are not easily etched

•e.g., Au

•Amenable to multilayer films

•Little or no substrate damage

•Generally low temperature process

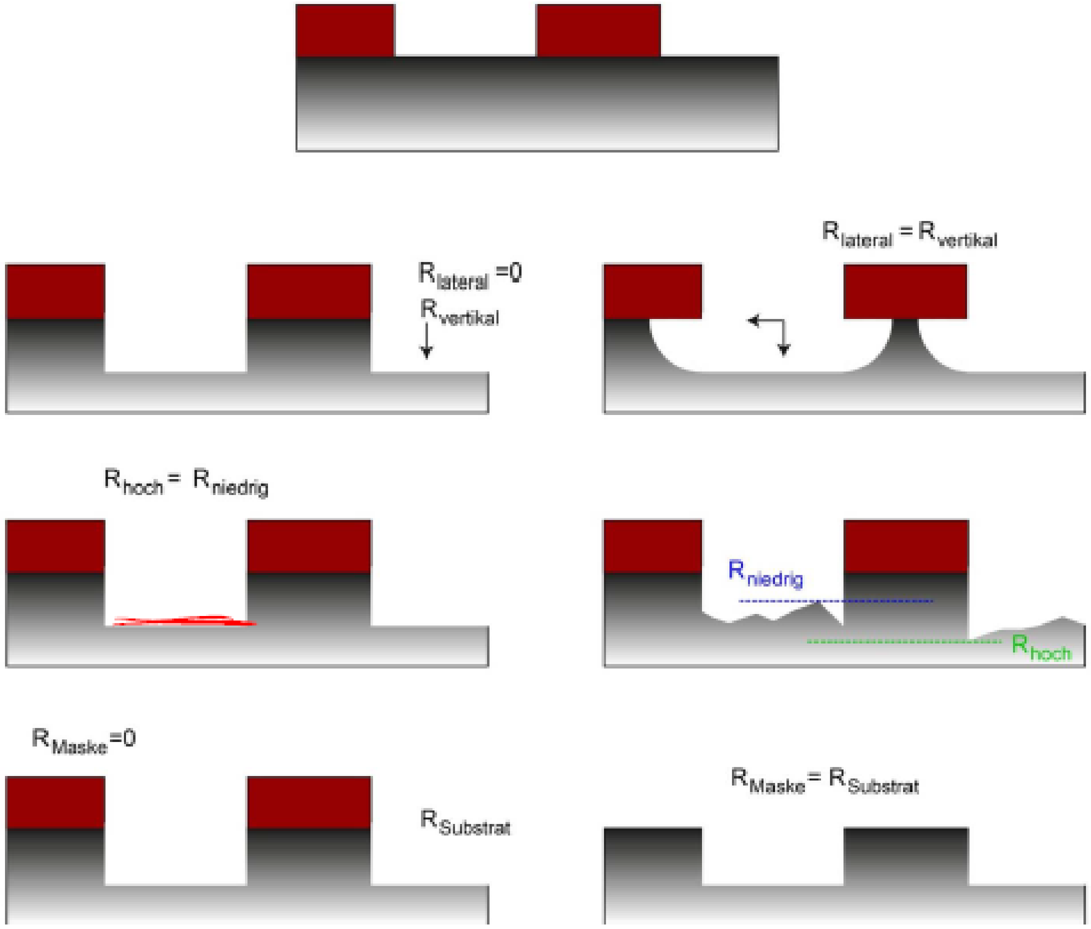
Outline

- 8. 1. Why is etching needed?
- 8. 2. Basics of etching
- 8. 3. Wet etching
- 8. 4. Dry etching
- 8. 5. Etching rate (dry)
- 8. 6. Summary and homework

Key factors in etching:

- **Uniformity**
- **Selectivity**
- **Anisotropy/Isotropy**
- Rate
- Damage

Key factors in etching:



anisotropy

$$A = \frac{R_{\text{vertical}} - R_{\text{lateral}}}{R_{\text{vertical}}}$$

uniformity

$$U = \frac{R_{\text{high}} - R_{\text{low}}}{R_{\text{high}} + R_{\text{low}}}$$

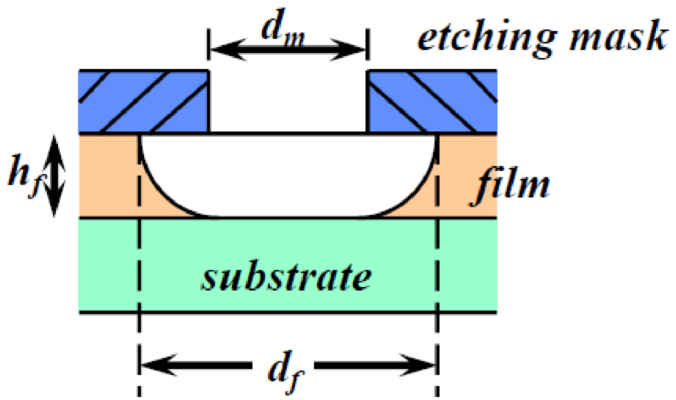
selectivity

$$S_{\text{m/s}} = \frac{R_{\text{mask}}}{R_{\text{substrate}}}$$

Uniformity of etching

(a): Film thickness variation across wafer

$$h_{f(max)} = h_f(1+\delta)$$
$$h_{f(min)} = h_f(1-\delta)$$



The variation factor δ is dictated by the deposition method, deposition equipment, and manufacturing practice.

(b) Film etching rate variation

$$\left.\begin{aligned} r_{f(max)} &= r_f(1+\delta) \\ r_{f(min)} &= r_f(1-\delta) \end{aligned}\right] \Rightarrow \left[\begin{aligned} t_{(max)} &= \frac{h_f(1+\delta)}{r_f(1-\delta)} \\ t_{(min)} &= \frac{h_f(1-\delta)}{r_f(1+\delta)} \end{aligned}\right.$$

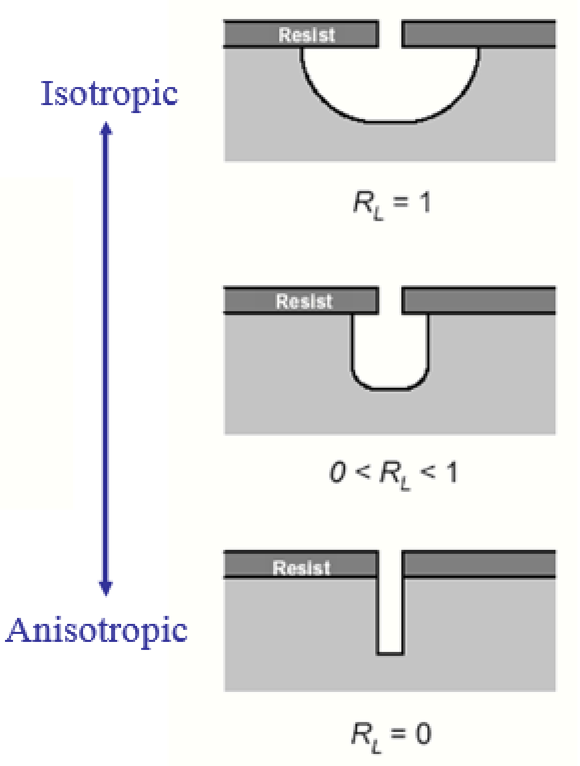
Directionality in Etching

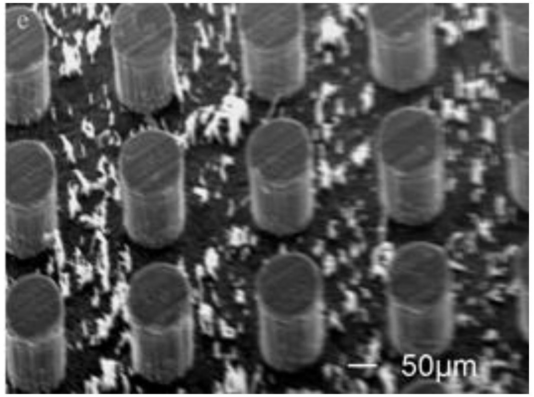
$$R_L = \frac{Horizontal\ \ Etch\ Rate\ (r_H)}{Vertical\ \ Etch\ Rate\ (r_V)}$$

Isotropic Etching: $R_L = 1$

Anisotropic Etching: $0 < R_L < 1$

Directional Etching: $R_L = 0$

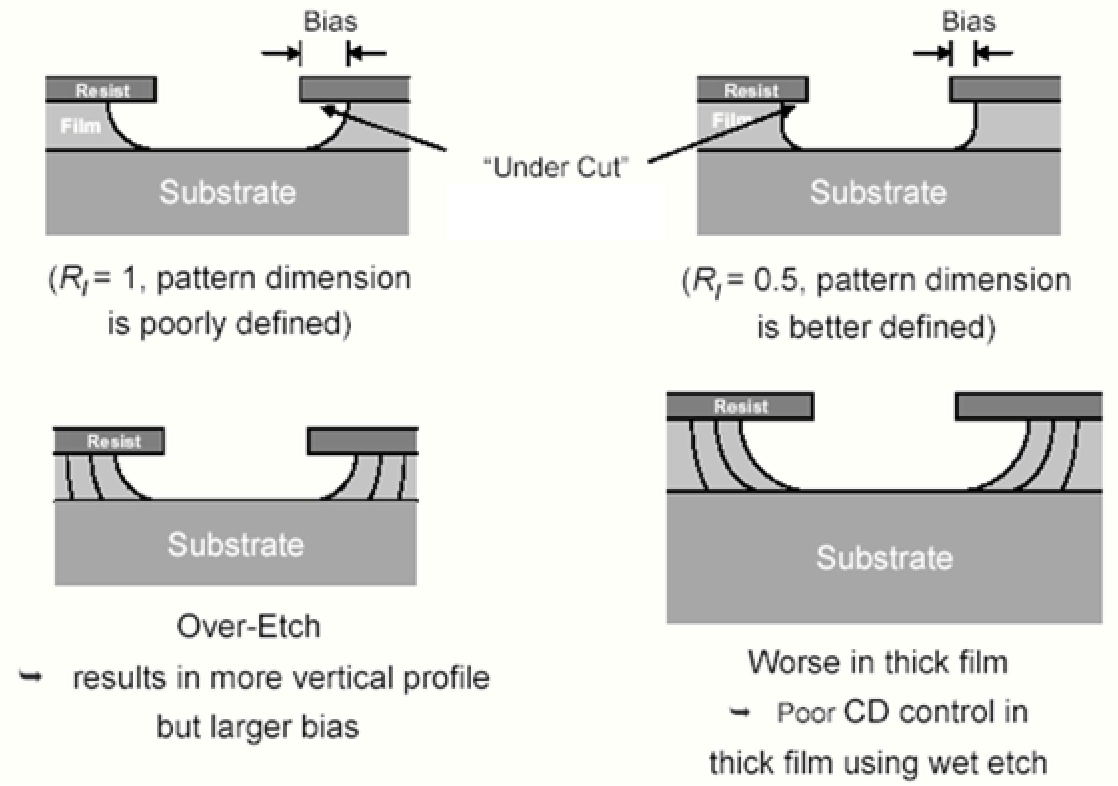


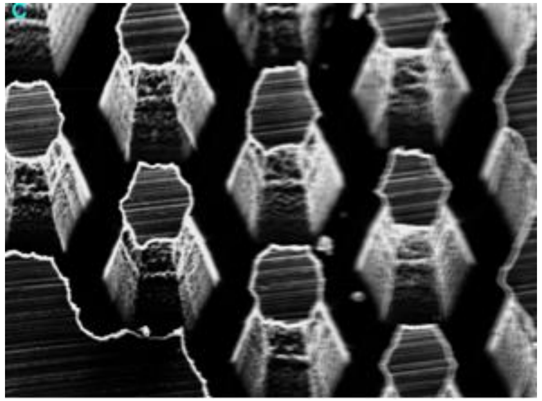


Etch Bias: Difference in lateral dimensions between resist features and etched pattern.

- smaller R_L = smaller bias

Undercut and Overetch





Requirements of etching

1. Obtain desired profile (sloped or vertical)
2. Minimal undercutting or bias
3. Large selectivity to other exposed films
4. Uniform and reproducible
5. Minimal damage to surface and circuit
6. Clean, economical and safe

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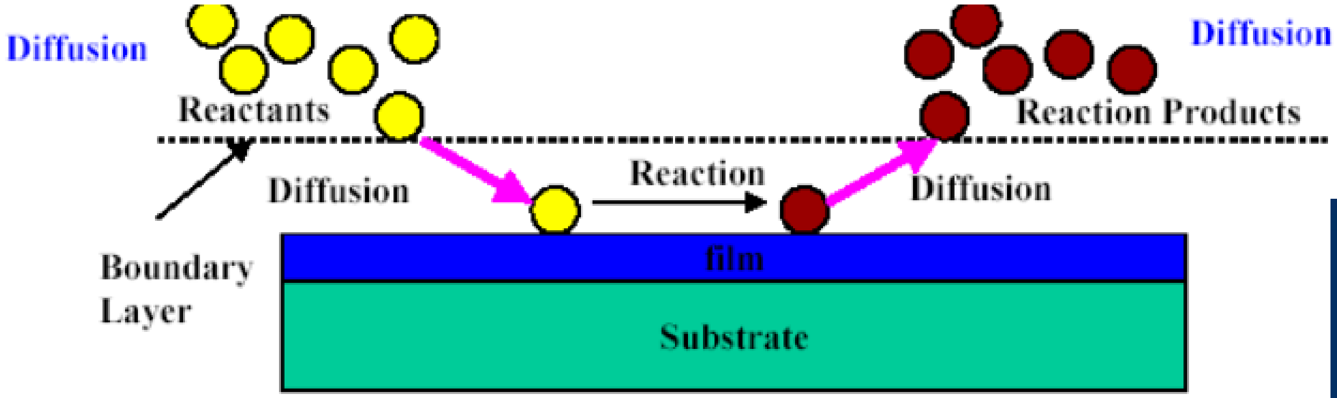
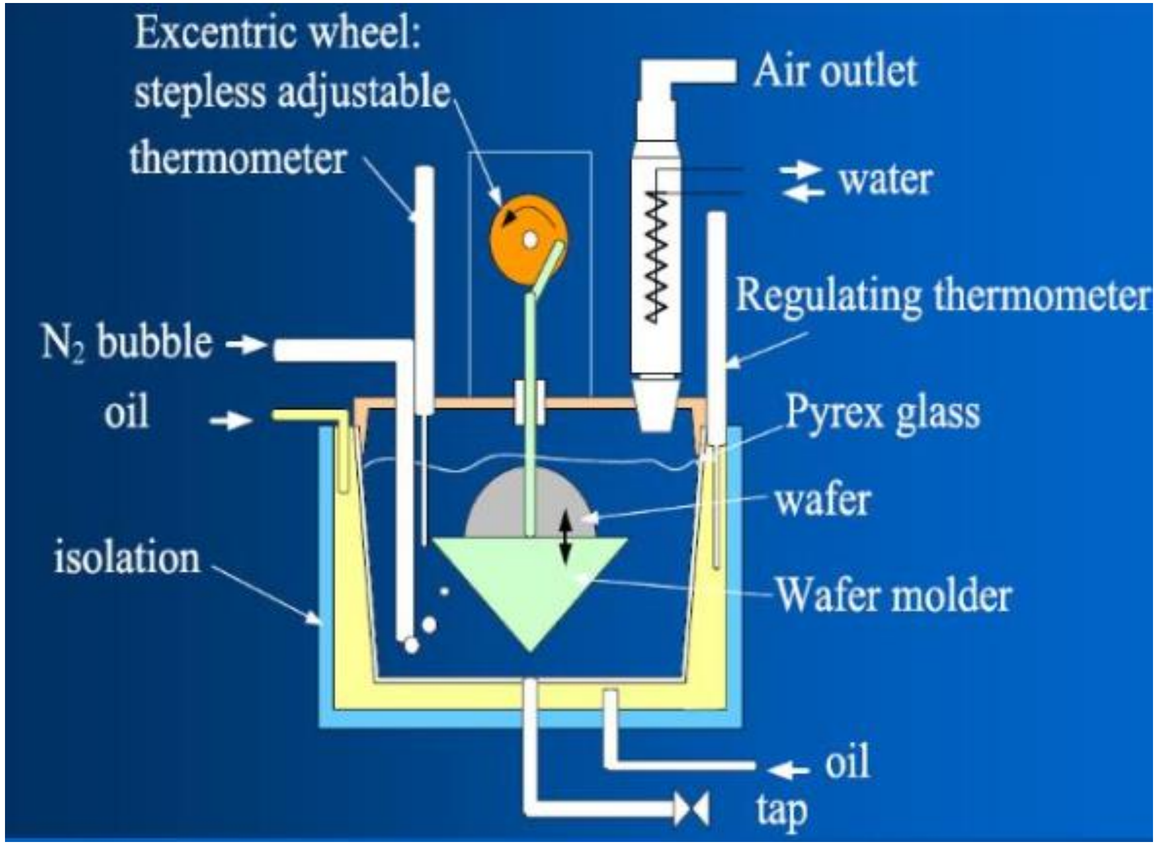
Principle of Wet Etching



反应产物必须溶于水或是气相



1. Reactant transport to surface
2. Selective and controlled reaction of etchant with the film to be etched
3. Transport of by-products away from surface

Characteristics of Wet Etching

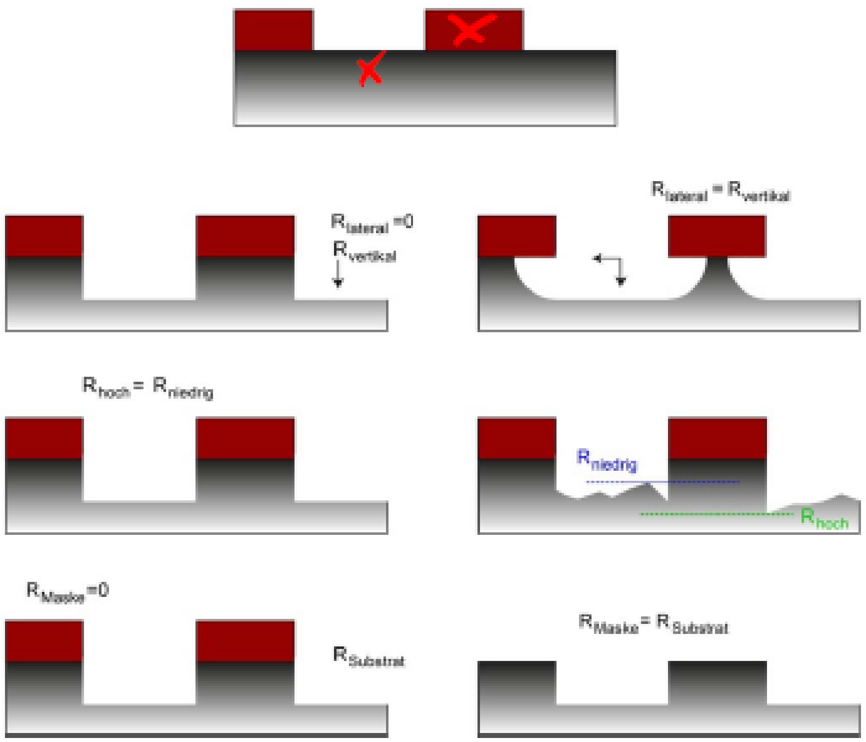
1) *Wet chemcial etching*

usually

- very high selectivity

- „simple“ and cheap

BUT:
- mostly isotropic etch behavior
- not very reproducible



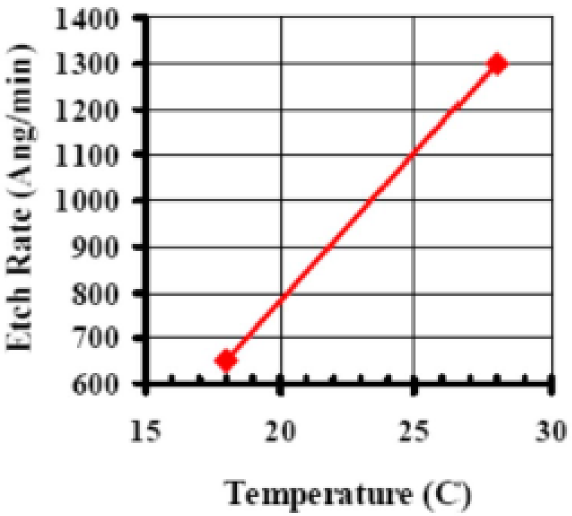
Wet Chemical Etching – SiO_2

To etch SiO_2 on Si, use HF solution

Note: HF is usually buffered with NH_4F to maintain $[H^+]$ at a constant level (for constant etch rate). This HF buffer is called Buffered Oxide Etch (BOE)

$$NH_4F \rightarrow NH_3 + HF$$

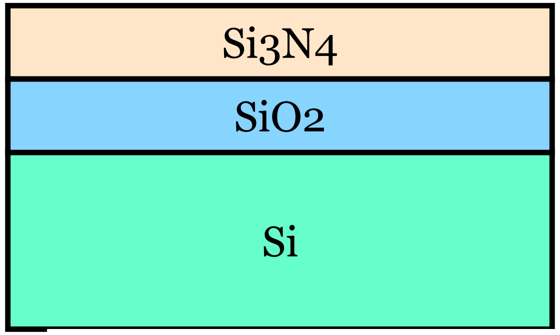
$$\boxed{SiO_2 + 6HF \rightarrow H_2 \cdot SiF_6 + H_2O \rightarrow H_2 + SiF_6 + 2H_2O}$$



- etch is isotropic and easily controlled by dilution of HF in water $6H_2O : 1HF$
- thermal oxides etches at
  - 120 nm/min in
  - 30 nm/s in 49% HF (by weight)
- doped/deposited oxide etches faster in HF
- selectivity to silicon $S_{SiO_2/Si} > 100$
- buffered HF (BHF) or buffered oxide etch (BOE) provides consistent etch rates
  - buffered with NH_4F

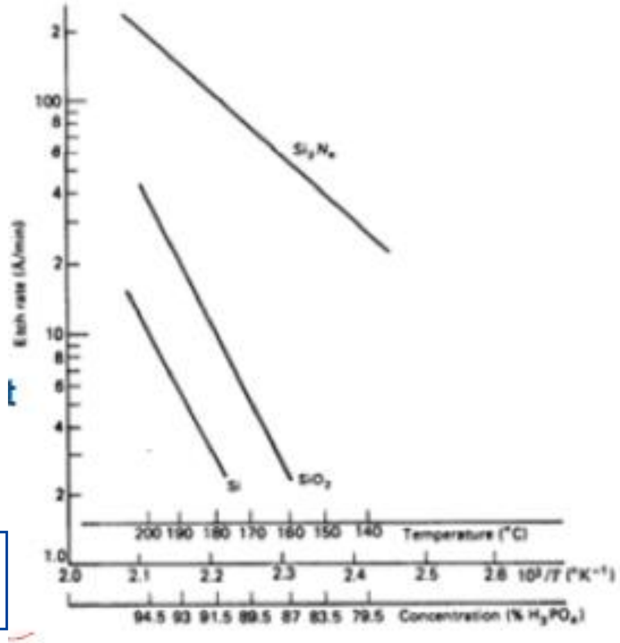
$$\boxed{6NH_4F : 1HF}$$

Wet Chemical Etching – Si_3N_4



- silicon nitride etched very slowly in **HF** solutions at room temperature
  - for example 20:1 BOE at RT
    thermal oxides etches at etch rate of SiO_2: **30nm/min**
    etch rate of Si_3N_4: **0.5-1.5nm/min**
  - very good selectivity of oxide to nitride
- silicon nitride etches in 49% HF at room temperature at approx. 50nm/min
- Phosphoric acid (**H_3PO_4**) at 150°C
  etch rate of Si_3N_4: 10nm/min
  etch rate of SiO_2: 1nm/min
  selectivity Si_3N_4 over SiO_2: Si=10 selelctivity of Si_3N_4 over Si: Si=30

So when you want to etch Si_3N_4 or SiO_2 what kind of mask can you select?



不同温度不同浓度的H_3PO_4

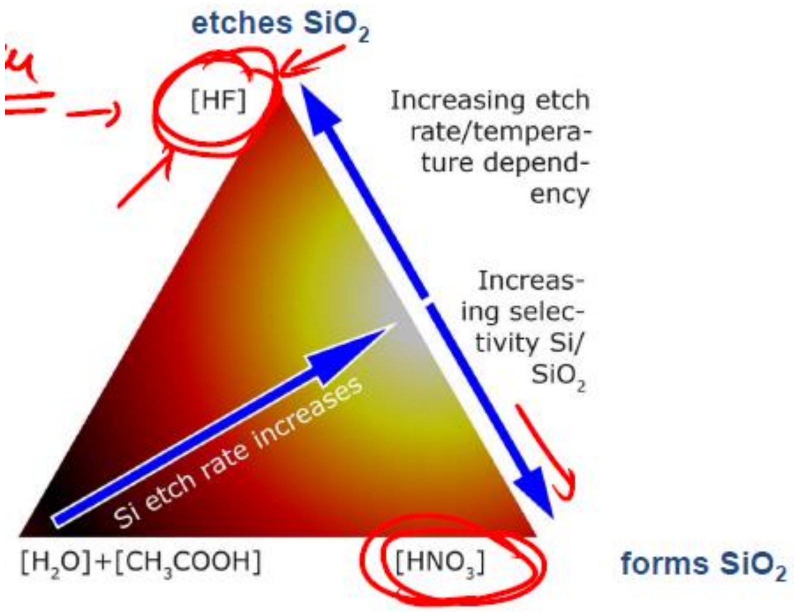
Wet Chemical Etching of Si

- isotropic etching

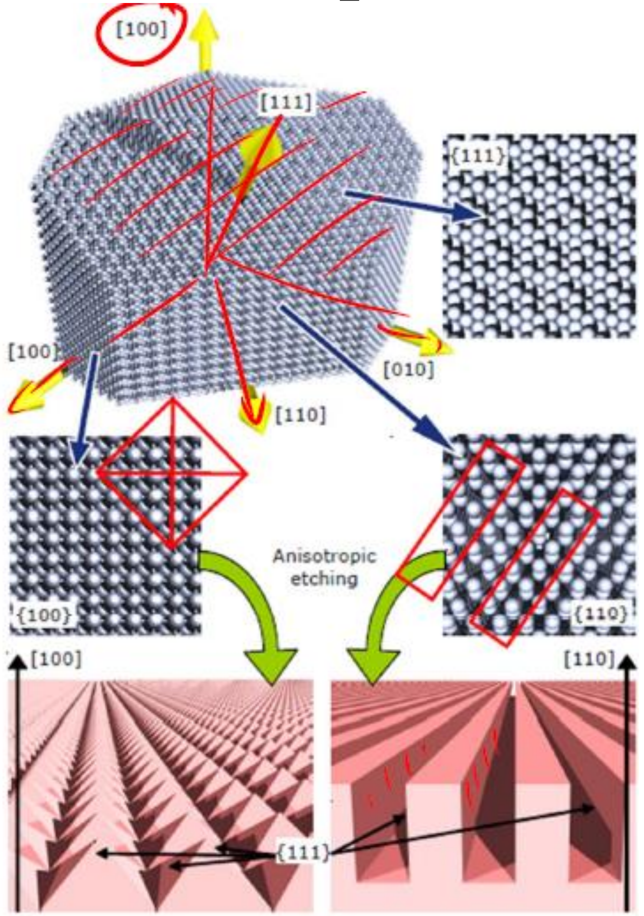
HNO_3 +HF + H_2O

(1) NO_2 formation (HNO_2 always in traces in HNO_3):	$HNO_2 + HNO_3$	$\rightarrow 2\ NO_2 + H_2O$
(2) Oxidation of silicon by NO_2:	$2\ NO_2 + Si$	$\rightarrow Si^{2+} + 2\ NO_2^-$
(3) Formation of SiO_2:	$Si^{2+} + 2\ (OH)^-$	$\rightarrow SiO_2 + H_2$
(4) Etching of SiO_2:	$SiO_2 + 6\ HF$	$\rightarrow H_2SiF_6 + 2H_2O$

improves wetting of hydrophobic surface/better uniformity

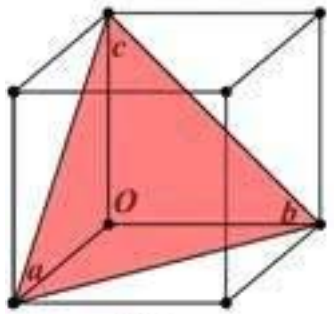


Anisotropic Wet Chemical Etching of Si



Etchant	Etch rate ratio		Etch rate (absolute)		
	(100)/(111)	(110)/(111)	(100)	Si_3N_4	SiO_2
KOH (44%, 85°C)	300	600	1.4 µm/min	<1 Å/min	14 Å/min
TMAH (25%, 80°C)	37	68	0.3-1 µm/min	<1 Å/min	2 Å/min

- **large difference of etch rate between different crystal directions**
- **etch results depends on wafer type (orientation)**
- **typical (100) substrates yields pyramidal structures, (110) oriented wafers: vertical grooves**



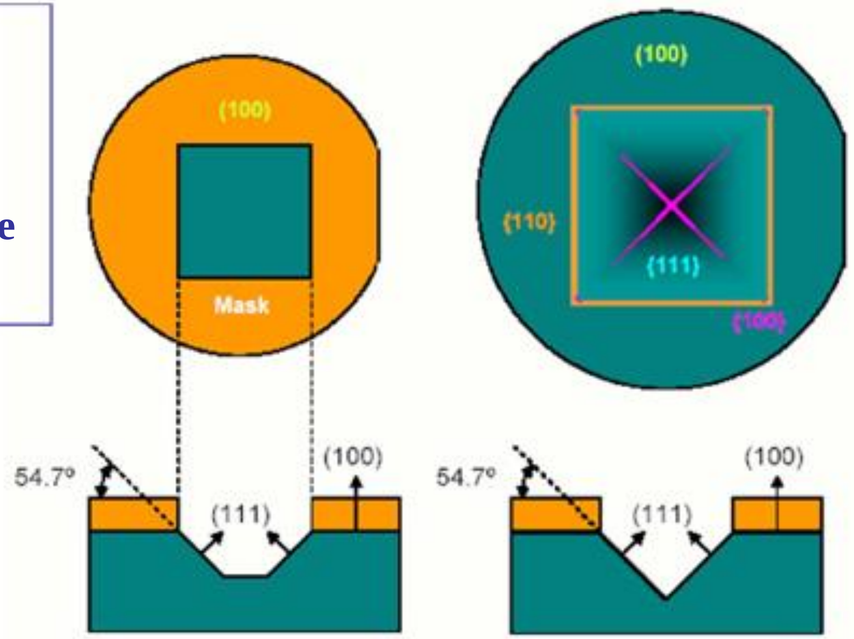
(111)

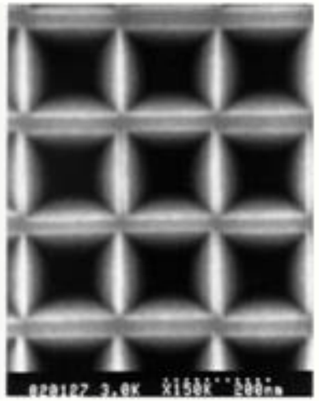
To understand the anisotropic etching of silicon, one should first understand the crystalline indices of silicon. The <111> direction has the highest density of atoms, next is the<110> direction and <100> is the lowest.

Anisotropic Wet Chemical Etching of silicon

Mask Aligned in <110> Direction

KOH
NaOH
LiOH
TMAH
Ethylenediamine
Hydrazine





200 nm pyramidal pits in Si (100)

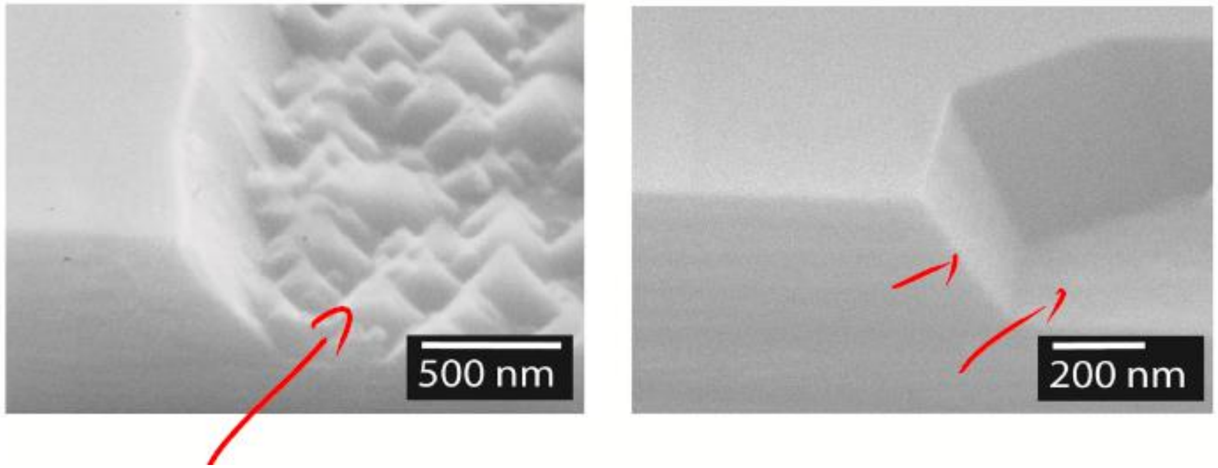
The rate of chemical erosion is highly dependent on the atom density of the crystal planes. The higher the atom density the lower the etch rate.

Surface atom density: {111} > {110} > {100}

Etch rate: $R_{(100)} \sim 100 \text{ X } R_{(111)}$

For the <100> oriented silicon wafer, etching through a square or rectangular mask opening will result in a square or rectangular opening in the <100> plane. This is not so for <110> plane

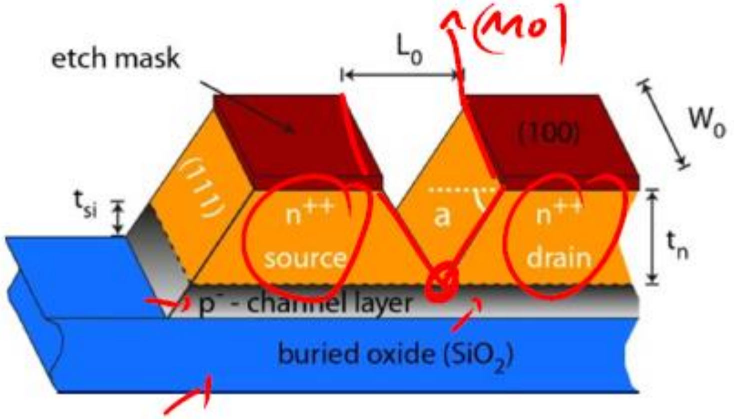
Anisotropic Wet Etching of Si

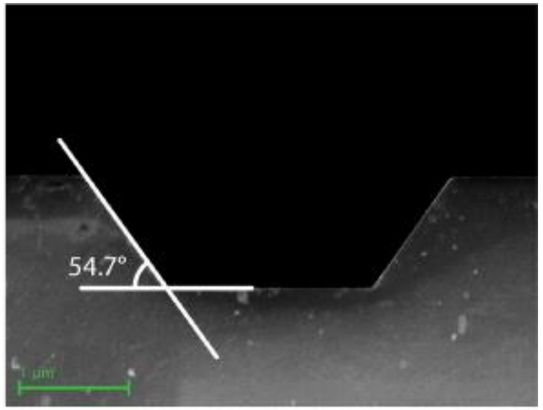


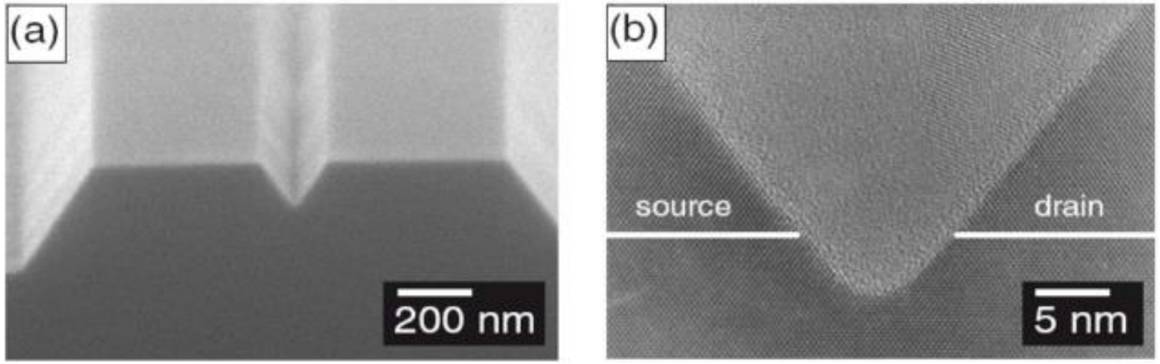
low KOH concentrations yield high etch rates but rough etch surface

high KOH concentrations yield smooth etch surfaces but low etch rate

Anisotropic Wet Etching of Si







Drawbacks of Wet Etching

- Lack of anisotropy
- Poor process control(**分辨率差**)
- Poor uniformity
- Excessive particulate contamination

Wet etching is used for noncritical feature sizes

Outline

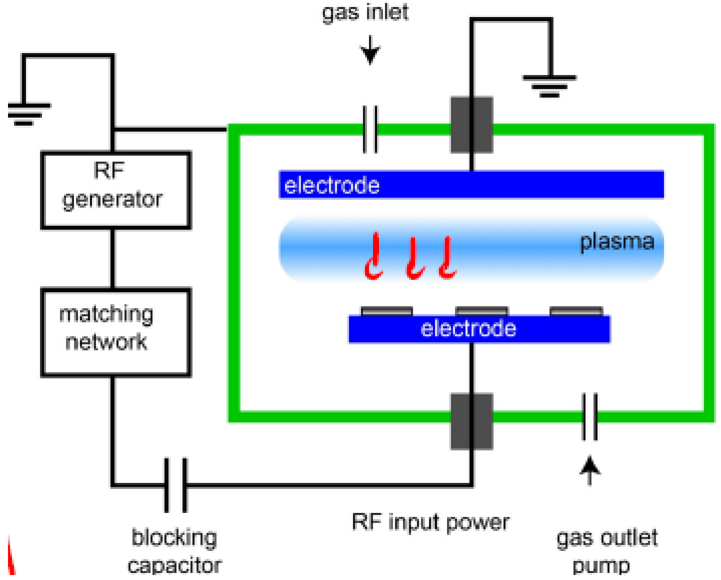
Dry etching

greater anisotropy than wet etching

Types of Dry Etching Technologies:

- Physical Sputtering(各向异性，选择性差)
  - Ion milling
  - Plasma sputtering
- Plasma-assisted chemical etching（选择性）

  eg, Reactive Ion Etching (RIE)

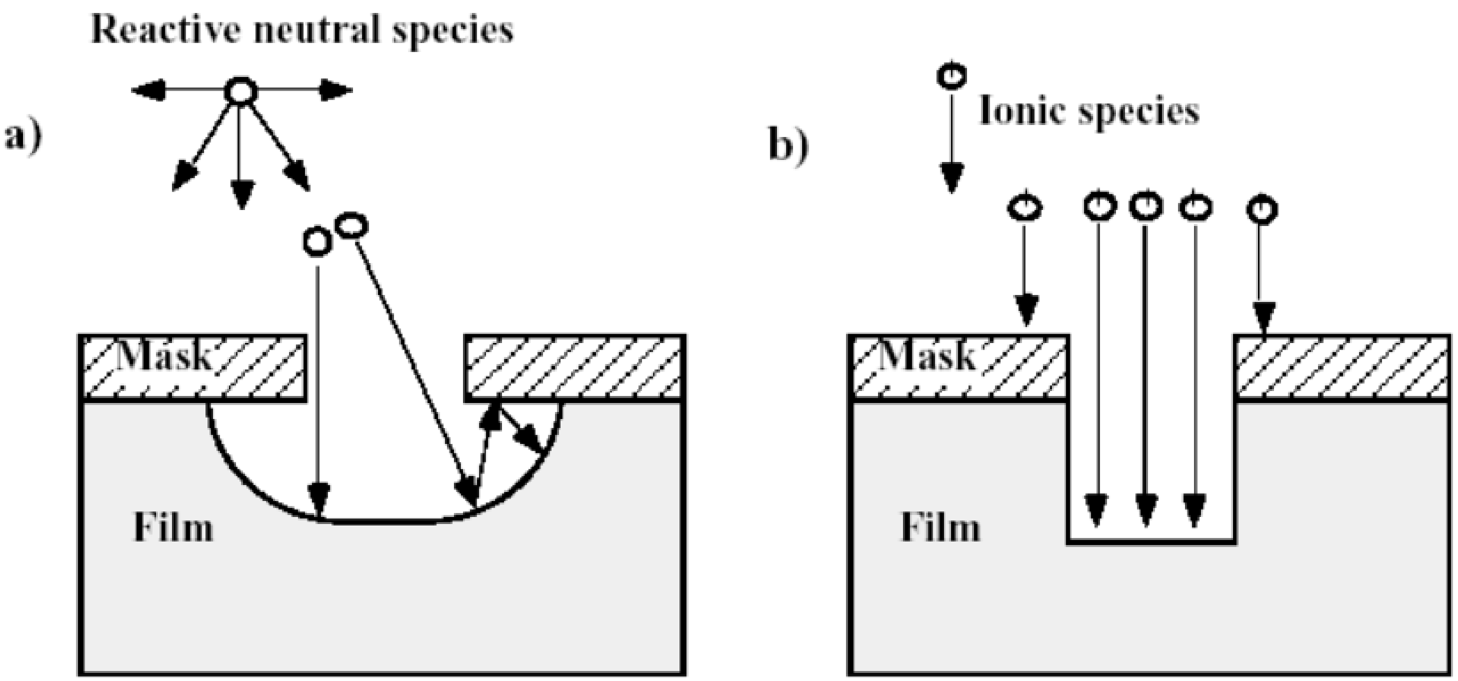


- Chemical etching + ion bombardment

Mechanism of plasma etching

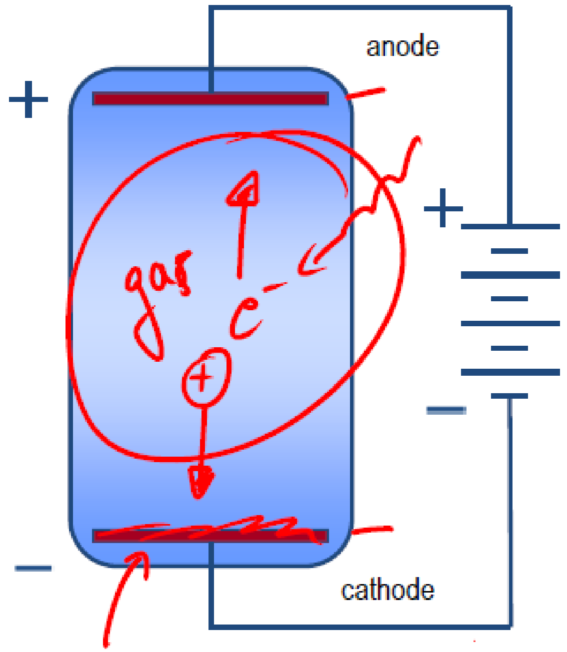
Three principal mechanism

- **Chemical etching (isotropic, selective)**
- **Physical etching (anisotropic, less selective)**
- **Ion-enchance etching/Reactive ion etching**



chemical etching physical etching

Dry Etching-plasma

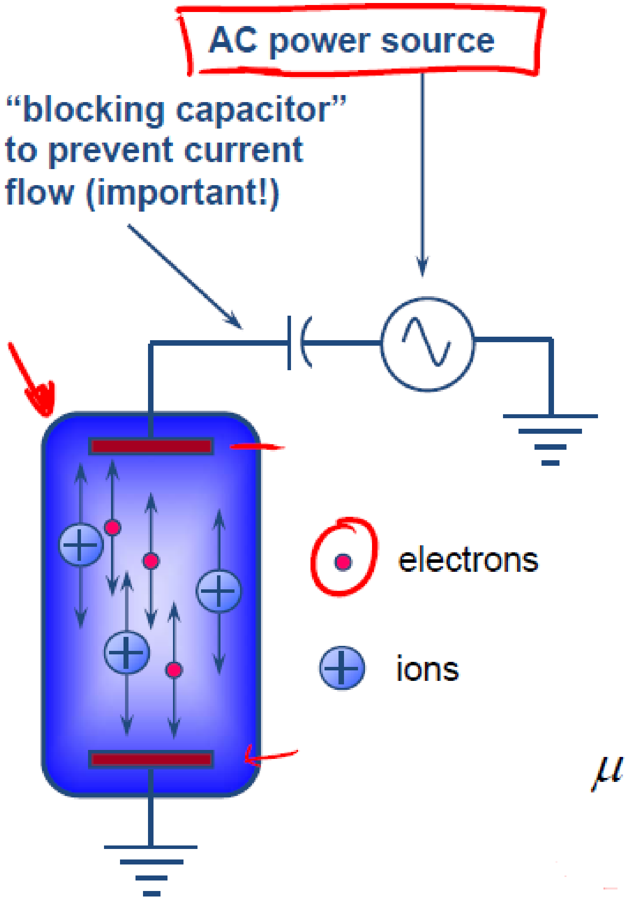


Cold plasma

simplest case: **parallel-plate-reactor**

- apply large voltage (kV-range) to electrodes
- both electrodes, cathode and anode need to be conductive
- gas feed in
- acceleration free electrons towards anode ionization of gas (hot electrons in cold gas)
- acceleration of ions towards cathode

Dry Etching- plasma

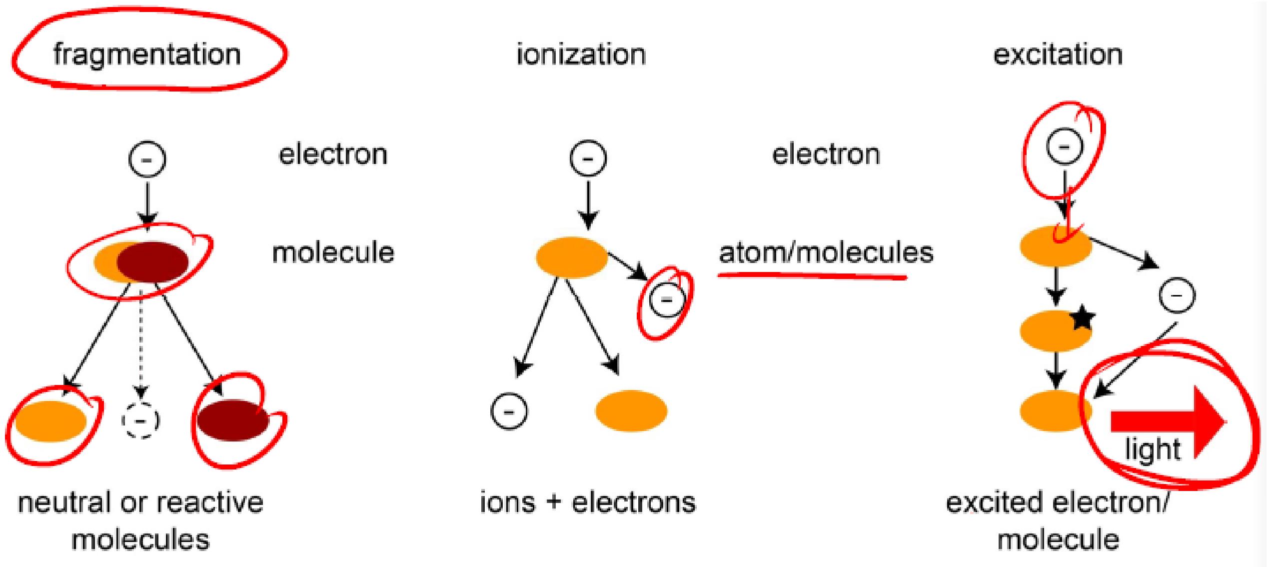


- in an AC discharge, an alternating potential is applied to one of the electrodes
- at low frequencies (~60 Hz) the relative polarities of the anode and cathode will merely alternate each ½ cycle
- at radio frequencies (13.56 MHz is the most common) major changes in the operation of the discharge will occur because of the different mobilities of electrons and ions:

$$\mu = \frac{e}{m\nu}$$

μ = mobility, *e*= electronic charge
m = mass, ν = collision frequency

Dry Etching-processes in plasma



plasma consists of 1% radicals and 0.01% ions

- **10^{15} cm^{-3} neutral**
- **10^{8} – 10^{12} cm^{-3} ions and electrons**
- **10^{12} – 10^{14} cm^{-3} reactive, neutral**
- **plasma density and ion-energy depend on each other**

Dry Etching- Reactive ion etching

Two dominating etch mechanisms:

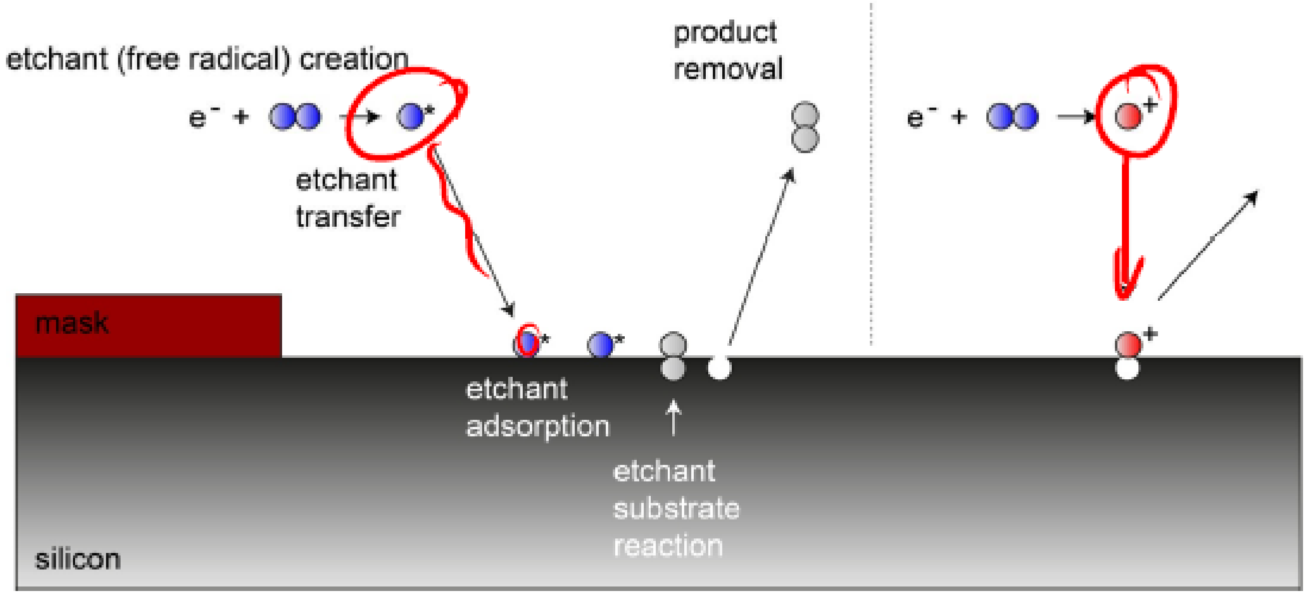
1. anisotropic / physical part: bombardement of surface with high-energy ions, yields sputtering of material

2. isotropic / chemical part: thermally activated, chemical etching process due to adsorption of neutral radicals, attacks vertical and horizontal surface: isotropic

E.g. Si-RIE: Silicon is etched with flourine or chlorine containing gases since this yields volatile compounds such as SiF_4. Process gases are for instance SF_6 , CF_4 , $SiCl_4$ oder BCl_3 eingesetzt.

Dry Etching- Reactive ion etching

Chemical component | physical component



Example: Si etching in CF_4 plasma

$$CF_4 \rightarrow F^\star + CF_3$$
$$CF_4 + e \longleftrightarrow CF_3^+ + F^\star + 2e$$
$$Si + 4F^\star \rightarrow SiF_4$$

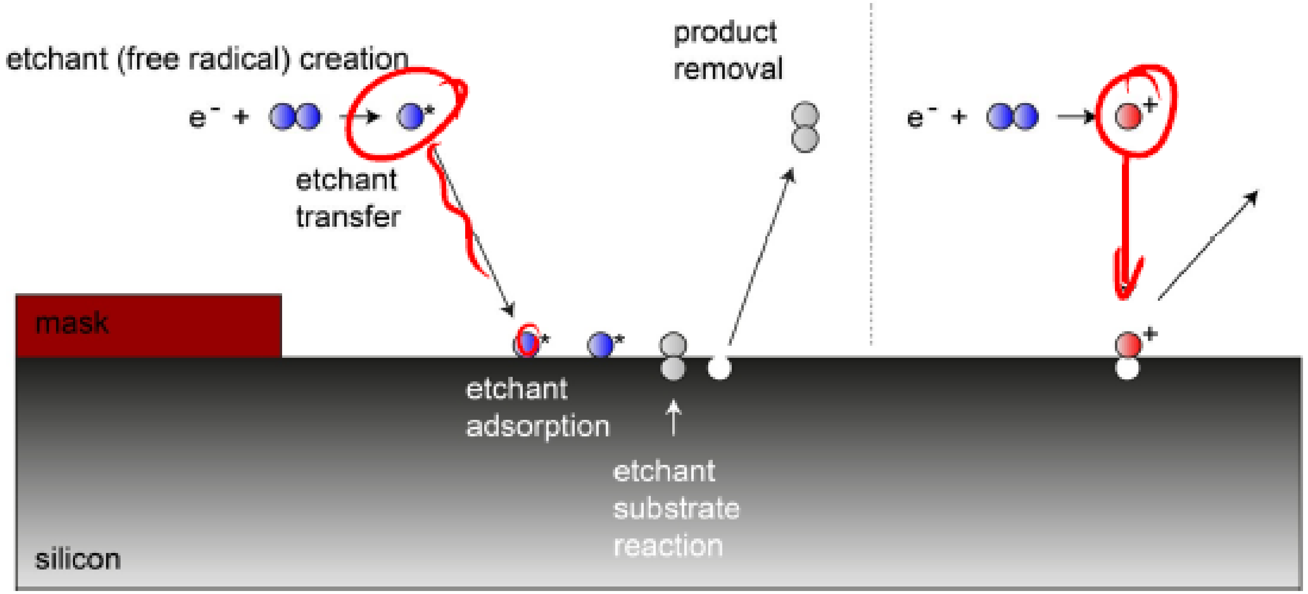
free radicals

SiF_4 is volatile, high vapor pressure

Additives like O_2 can be used which react with CF_3 and reduce CF_3+F recombination. Higher etch rate

Dry Etching- Reactive ion etching

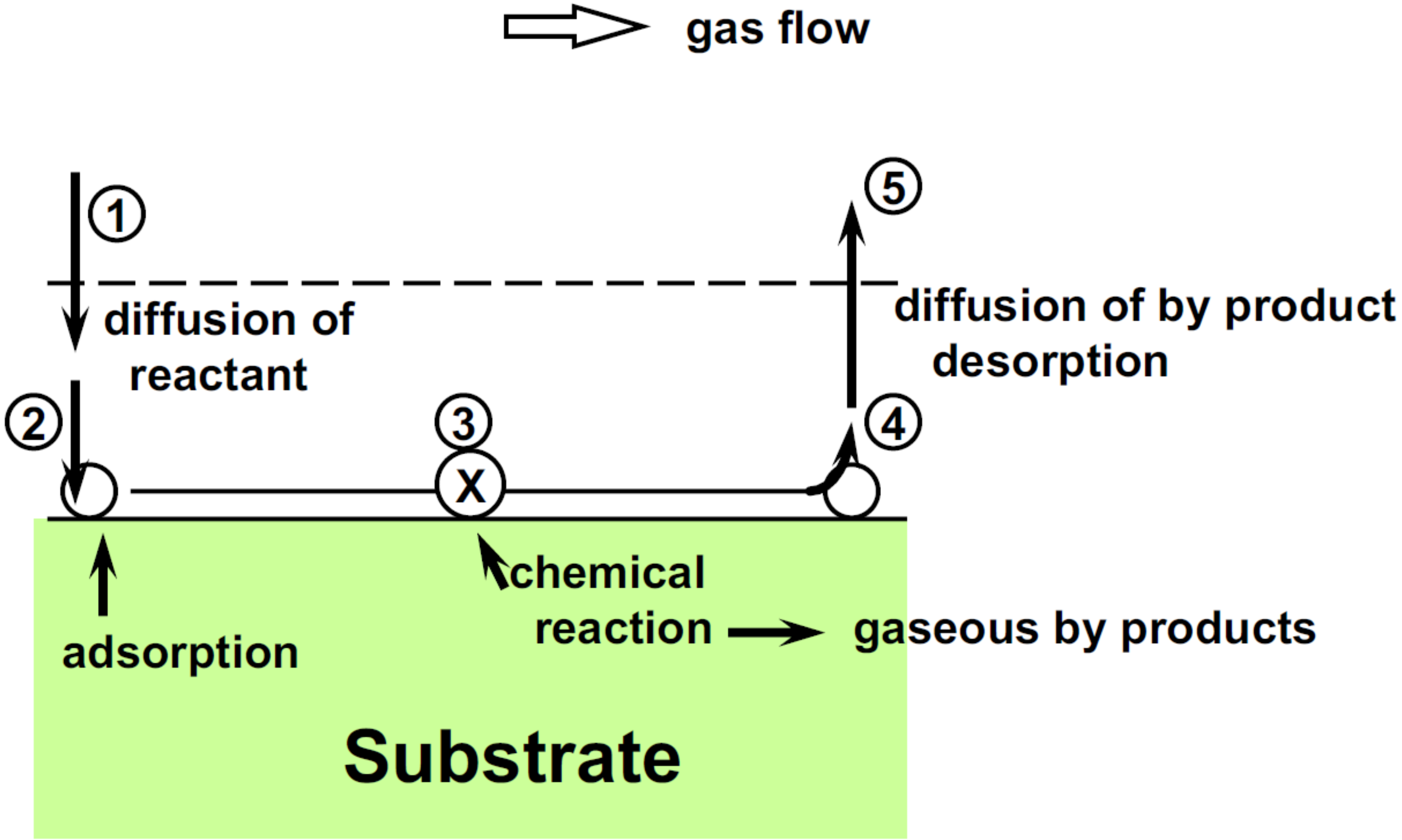
Chemical component physical component



Example: Resist ashing in oxygen plasma

$C_xH_yO_z + O_2 \rightarrow CO_x + HO_x$ **volatile carbon oxide compounds**

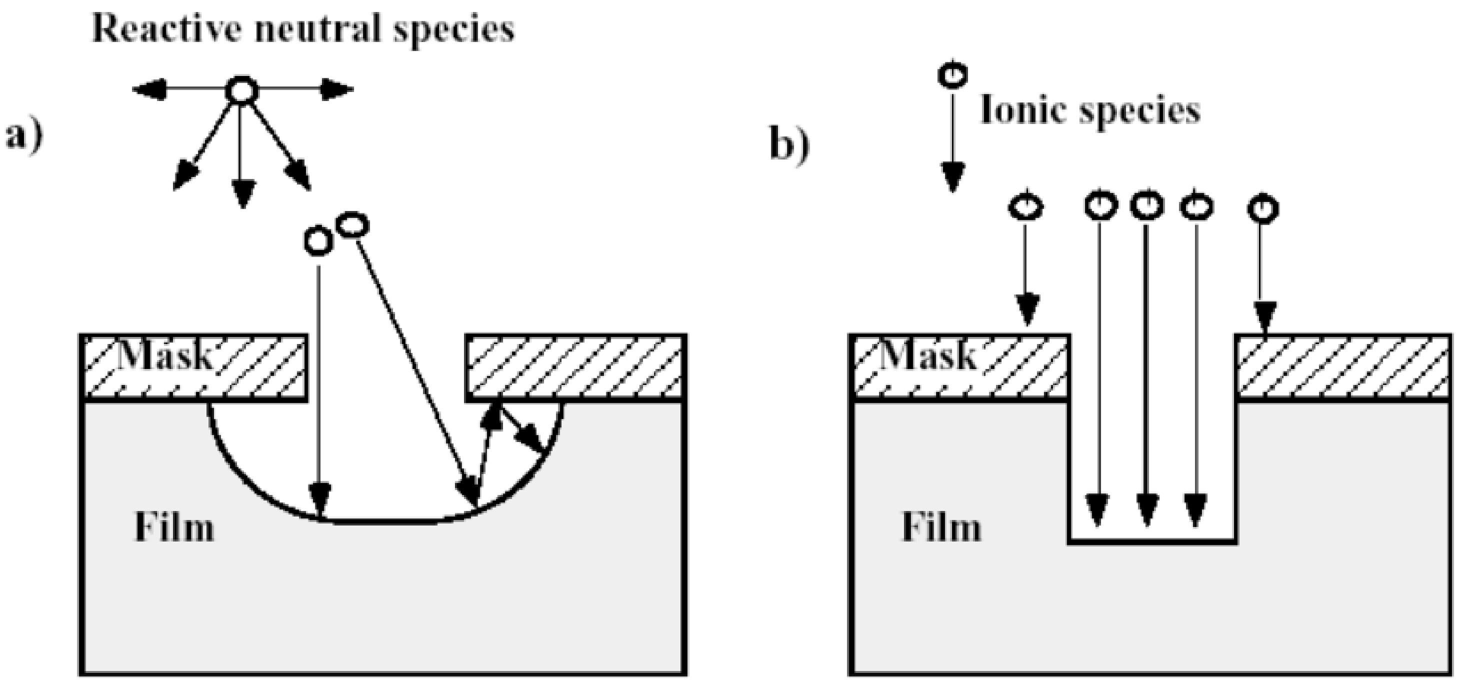
Etching sequence of chemical etching



Physical etching

Etching species are high energy ions like CF^{4+} or Ar^{+} which remove material by sputtering.

- Not very selective but more directional.
- Can cause damage to surface, with extent and amount of damage a direct function of ion energy.

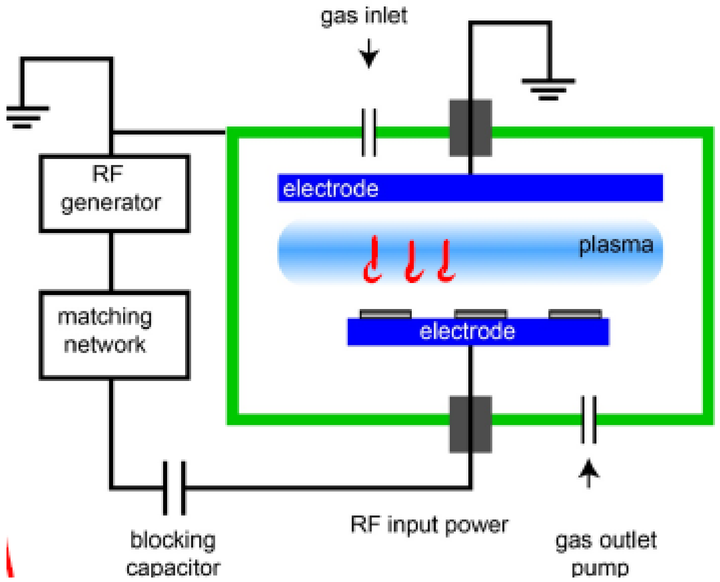


chemical etching

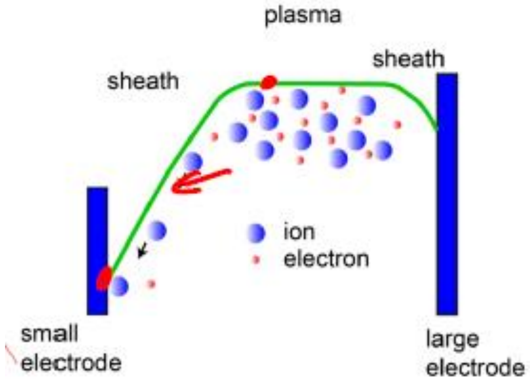
physical etching

Dry Etching- Reactive ion etching

- high electric field at wafer-electrode, DC bias voltage approximately 100-700V
- ions strongly accelerated yields anisotropic, vertical etching
- physical *and* chemical etching components



- anisotropic etching and little undercutting because of direct ion flux
- selectivity due to chemical component



problem:
plasma density and Ion energy are related to each other

Dry Etching- Reactive ion etching

Issue: parallel-plate-reactor: plasma density and ion-energy depend on each other

Higher plasma density requires more RF power → yields higher ion energy, reduces selectivity, causes damages

ECR- RIE

Electron-Cyclotron-Resonance Reactive Ion Etching

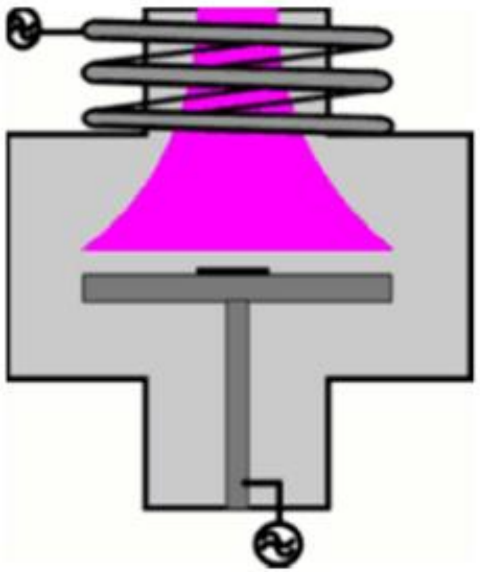
ICP- RIE

Inductively-Coupled Plasma Reactive Ion Etching

ICP RIE

Key features

- Separate RF generators for ICP and electrode
  - Separate control over ion energy and ion density
  - Eliminates capacitive coupling
  - High ion density ($>10^{11}$ cm^{-3})
    - (RIE: ~ 10^{9} cm^{-3})
- Low ion energy possible
- Separate control over ICP and electrode RF

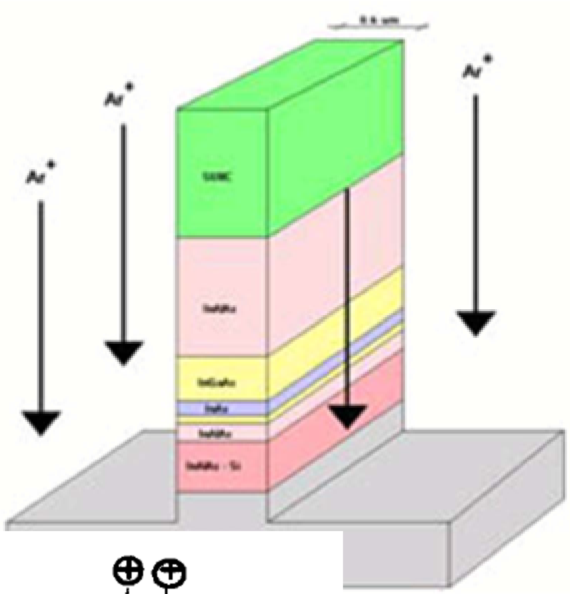


Benefits

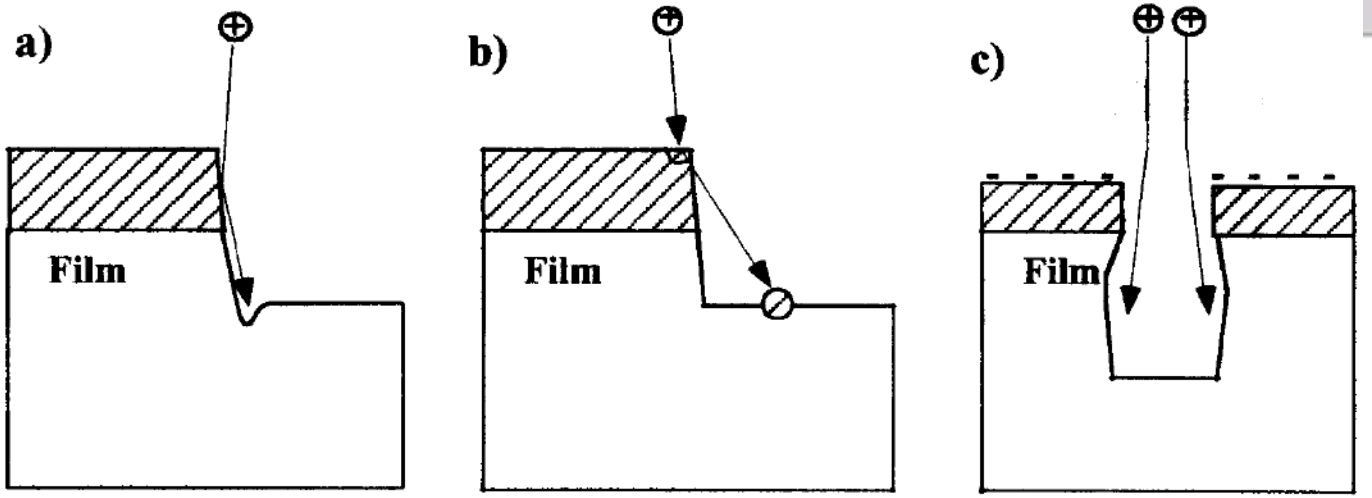
- High etch rates
- Control over selectivity and damage
- High process flexibility
- Chemical and ion-induced etching
  - Reduced electrical damage to devices
  - Reduced chamber particles

Ion Milling (Sputter Etching)

- Purely physical etching
- Sputter etching using Ar+
- Damage to wafer surface and devices



Material	Etch Rate in Ar+ (Å/min.)	
	E = 175 eV	E = 475 eV
Ta	10	165
Cu	15	285
NiFe	20	150
PtMn	25	140
Al_2O_3	4	32
Au	95	500
Cr	25	115
SiO_2	5	125

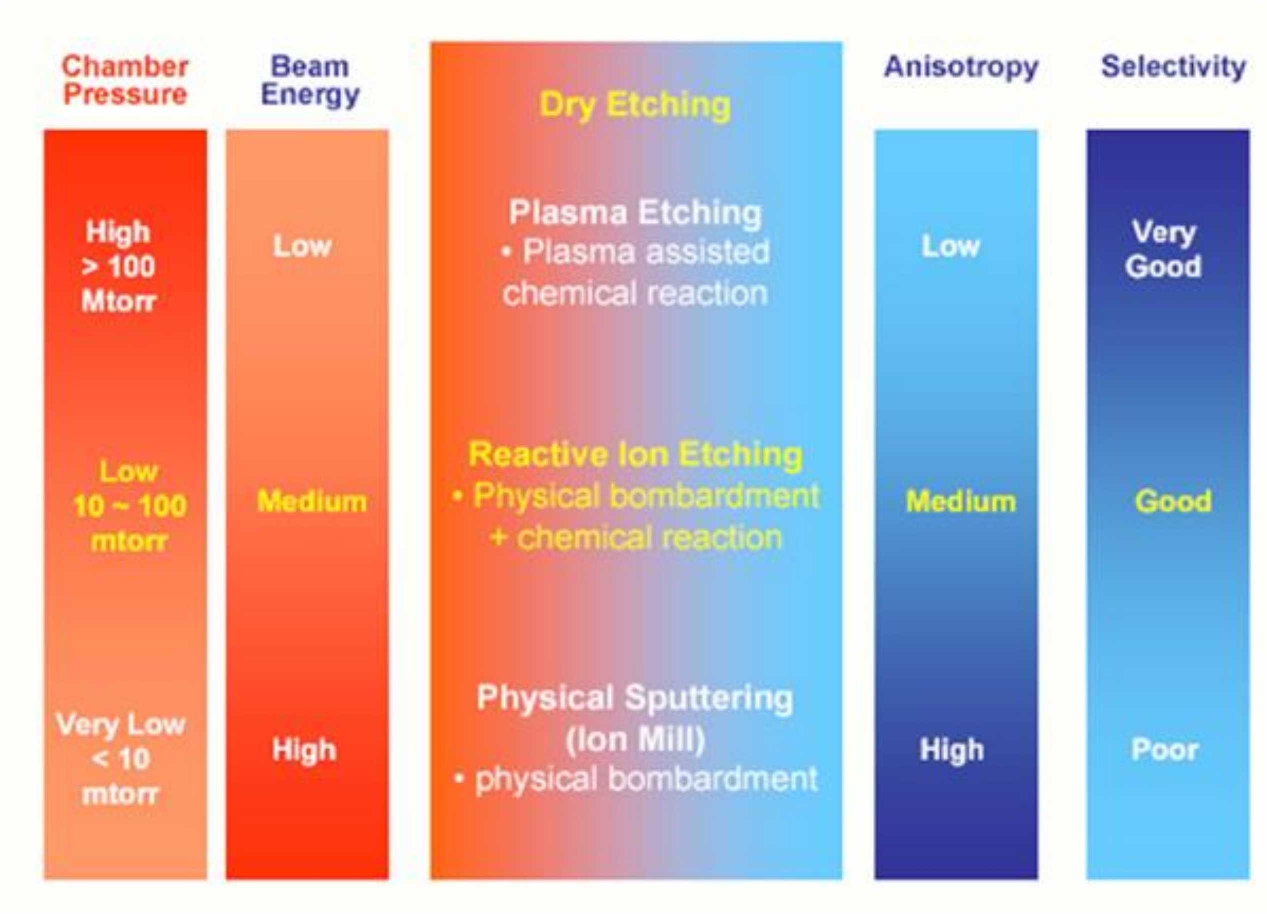


trenching

redeposition

charging and ion path distortion

Dry Etching Comparison



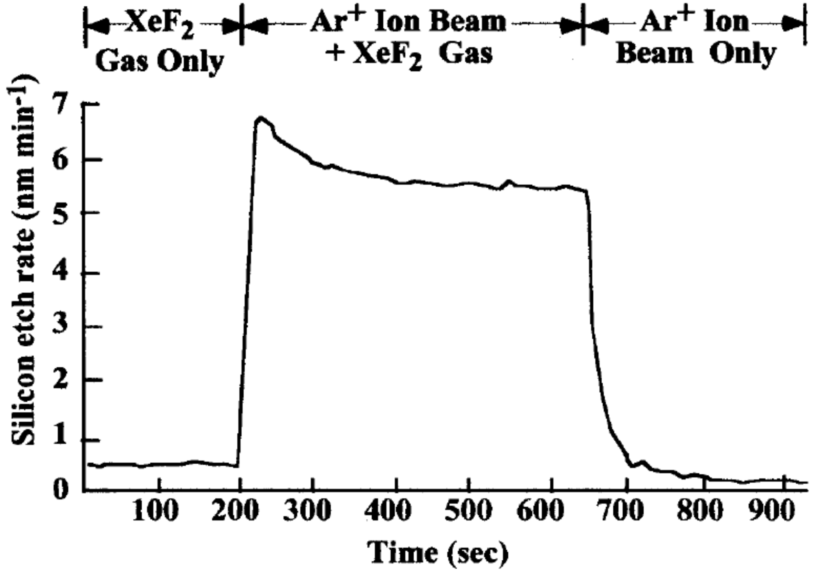
Outline

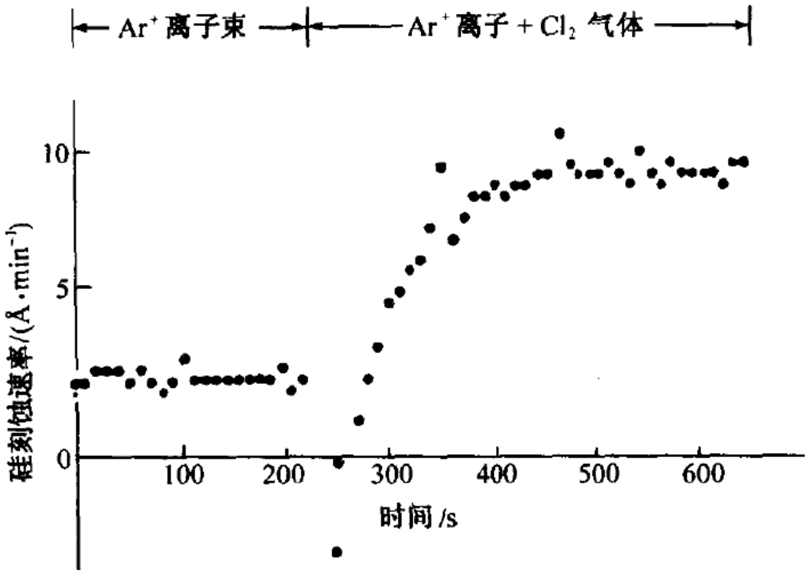
Influence factors

- Etching method (chemical, physical and RIE)
- Gas compositions
- Gas pressure and flow rates
- RF power
- Other factors

Dry Etching rate- Reactive ion etching

Two kinds of mechanism for enhanced etching rate

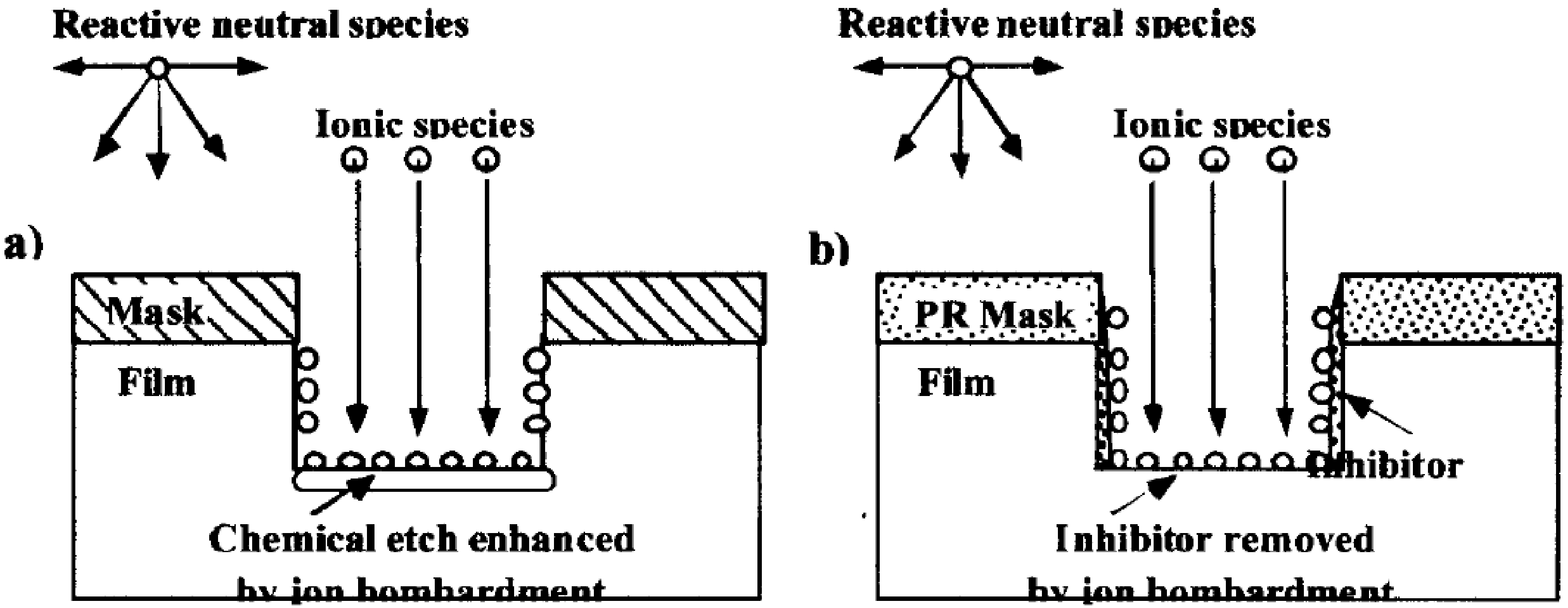




- Enhanced etching is observed when both kinds of etching is involved in some cases.

Dry Etching rate- Reactive ion etching

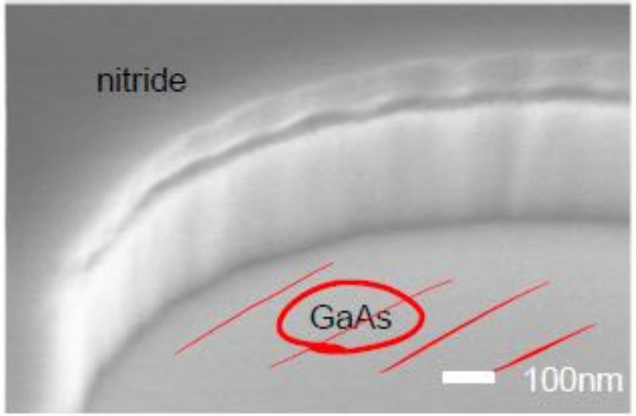
Two kinds of mechanism for enhanced etching rate.

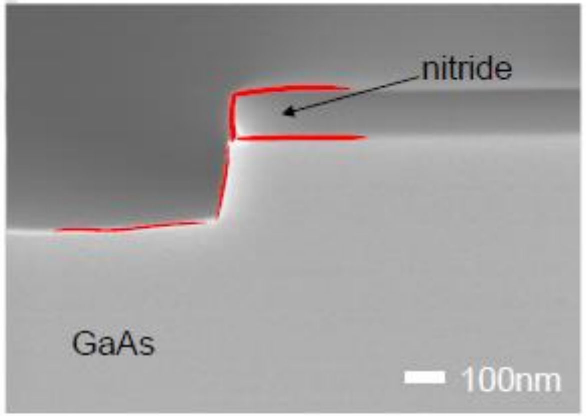


Effect of gas composition

Material	Gas	Mask	Selectivity
Si (a-Si)	1) CF4 2) SF6 3) BCl2 + Cl2	Resist Metal (Cr, Ni, Al)	~ 20:1 ~ 40:1
SiO_2	1) CHF3 + O2 2) CF4 + H2	Resist Metal (Cr, Ni, Al)	~ 10:1 ~ 30:1
Si_3N_4	1) CF4 + O2 (H2) 2) CHF3	Resist Metal (Cr, Ni, Al)	~ 10:1 ~ 20:1
GaAs	1) Cl2 2) Cl2 + BCl3	Si_3N_4 Metal (Cr, Ni)	~ 10:1 ~ 20:1
InP	1) CH4/H2	Si_3N_4 Metal (Cr, Ni, Al)	~ 40:1
Al	1) Cl2 2) BCl3 + Cl2	Resist Si_3N_4	~ 10:1
Resist / Polymer	1) O2	Si_3N_4 Metal (Cr, Ni)	~ 50:1

Dry Etching- Reactive ion etching/anisotropy

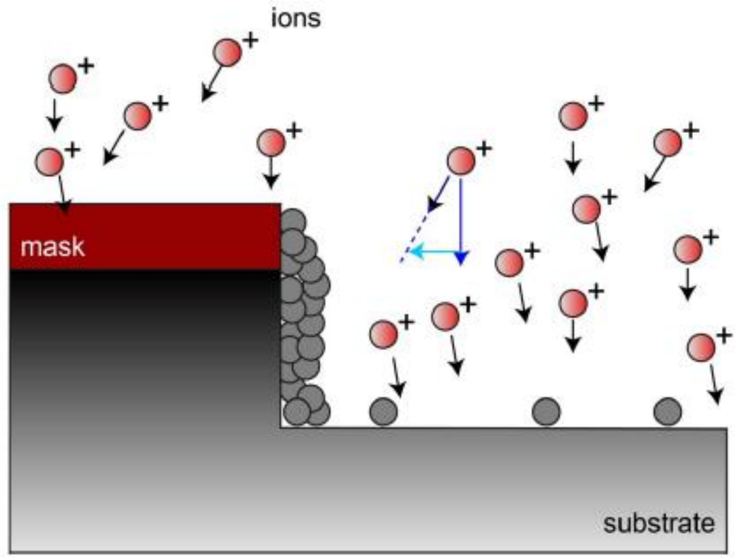




dry (ECR-RIE) mesa etching with ECR-PECVD nitride as etch mask (2” GaAs)

Gases: CH_4 / H_2

Dry Etching- Reactive ion etching/anisotropy



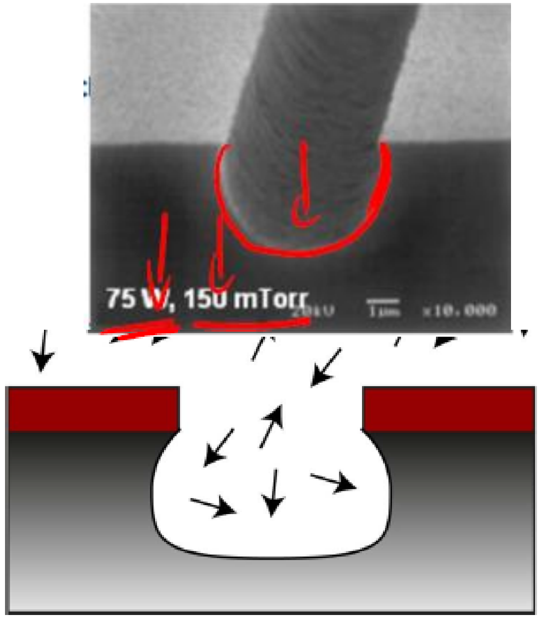
increase of anisotropy due to polymer-deposition(passivation)

physical component due to vertical ion bombardment removes polymer on horizontal surfaces

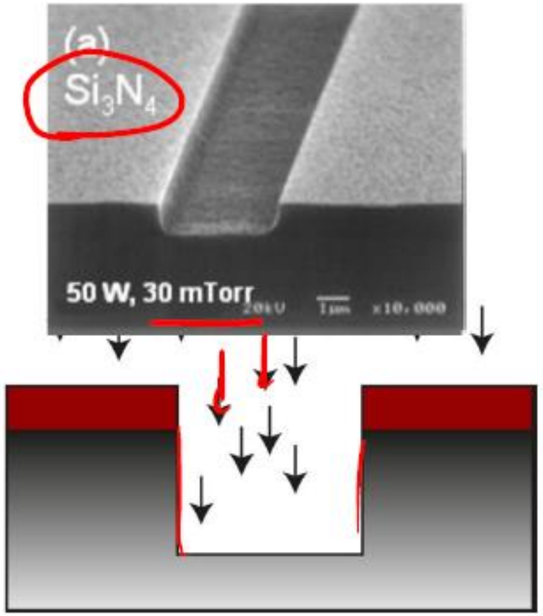
side-walls are passivated leading to anisotropic etching

Example:
- Silizum etching in CF_4/H_2
- hydrogen-methane process for etching III-V semiconductors

Dry Etching- Reactive ion etching/anisotropy



rather isotropic etch behavior



anisotropic etch behavior

large mean free path of ions at low chamber pressure: vertical bombardment of sample surface leads to anisotropic etching (along field lines)

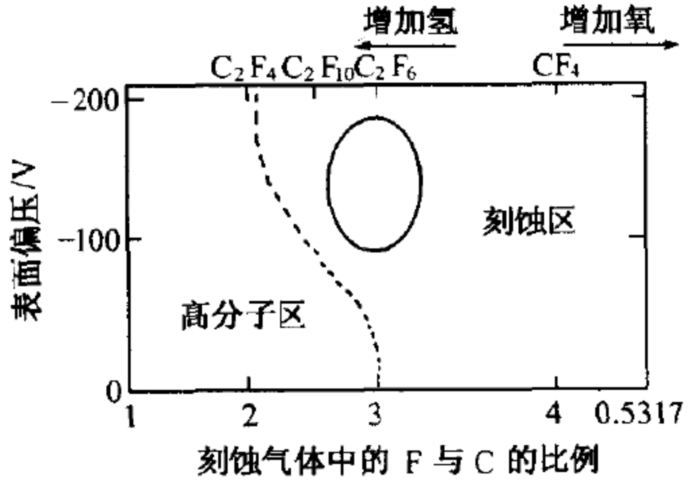
Drying etching for SiO_2 or Si

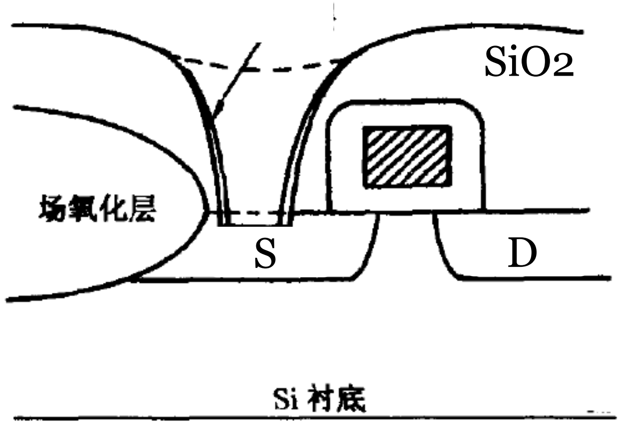
$$CF_4 + e \longrightarrow CF_3 + F(\text{游离基}) + e$$

$$SiO_2 + 4F \longrightarrow SiF_4(\text{气}) + O_2$$

$$Si + 4F \longrightarrow SiF_4(\text{气})$$

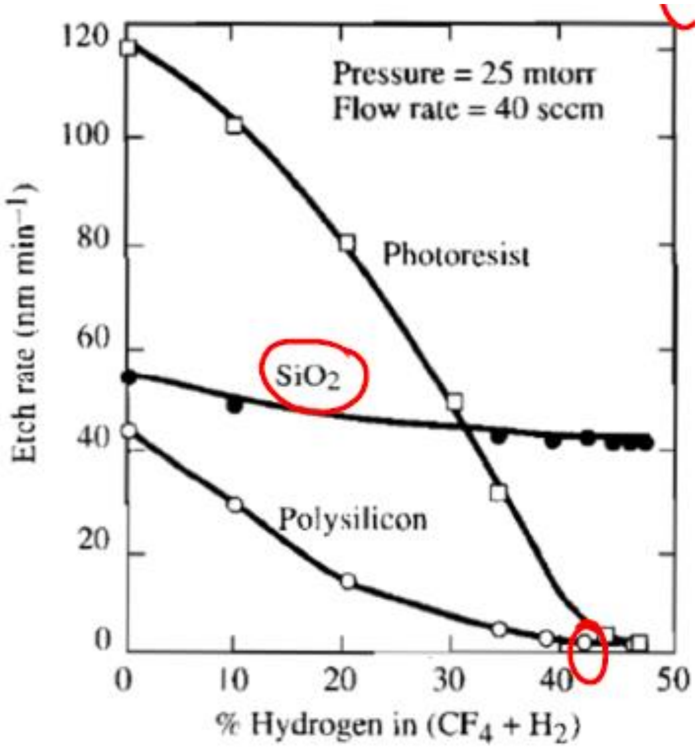
Additives like O_2 can be used to react with CF_3 and increase high etching rate. And additive like H_2 can be used to reduce F/C ratio to reduce etching rate





Dry Etching- Reactive ion etching/selectivity

Selectivity during silicon etching: CF_4 / H_2 gas mixture



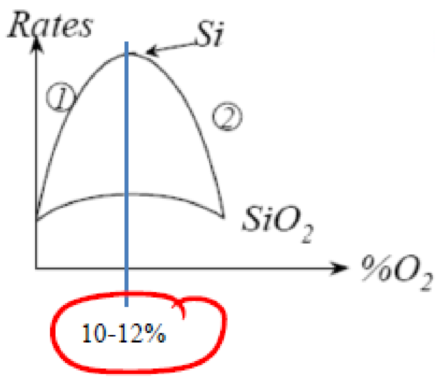
adding H_2 drastically lowers Si etch rate
- H_2 lowers F concentration (H^+ +F+ e^- HF)
- etch rate nearly zero at 40% H_2

- etch rate of SiO2 remains nearly constant (HF!)

- allows etch selectivity to be increased tremendously

Dry Etching- Reactive ion etching/selectivity

Selective etching of Si/SiO2 with CF_4/O_2



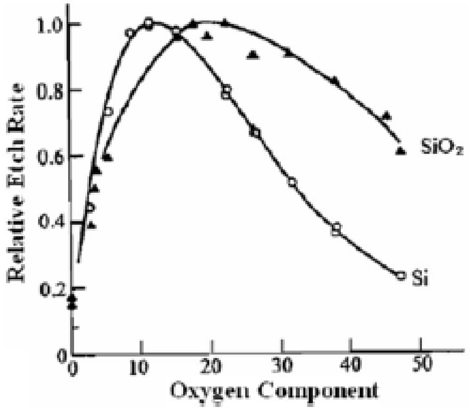
$$O + CF_x \rightarrow COF_x + F^\star$$

$F_\star$ increases Si etch rate

$$Si + O_2 \rightarrow SiO_2$$

SiO_2 etch rate decreases

adding O_2 or H_2 allows adjusting selectivity for etching Si with respect to SiO2

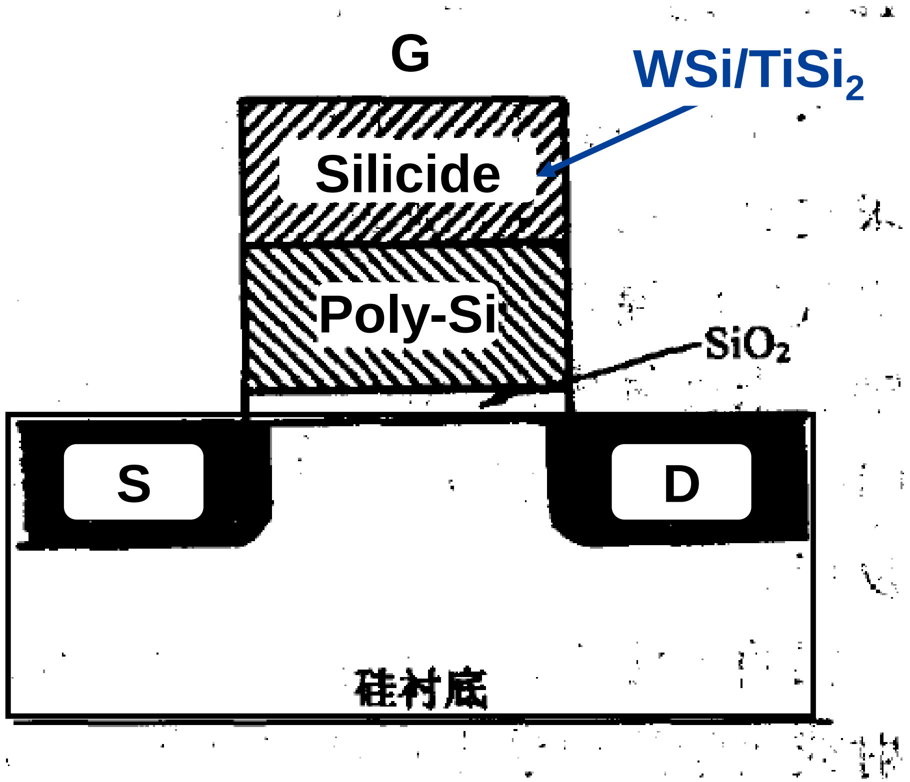


Drying etching for poly Si and MSix

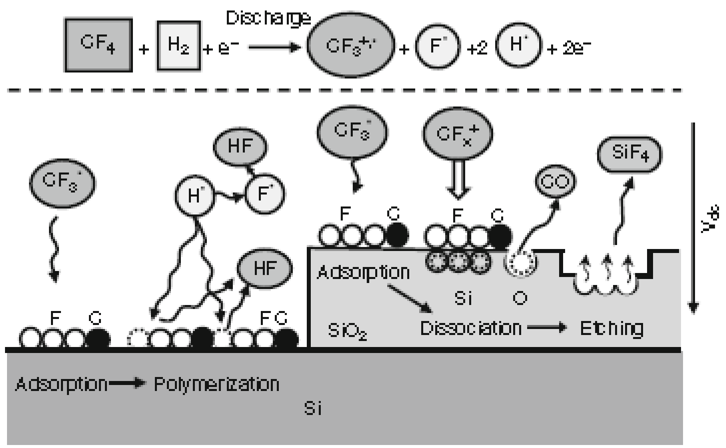
Two steps

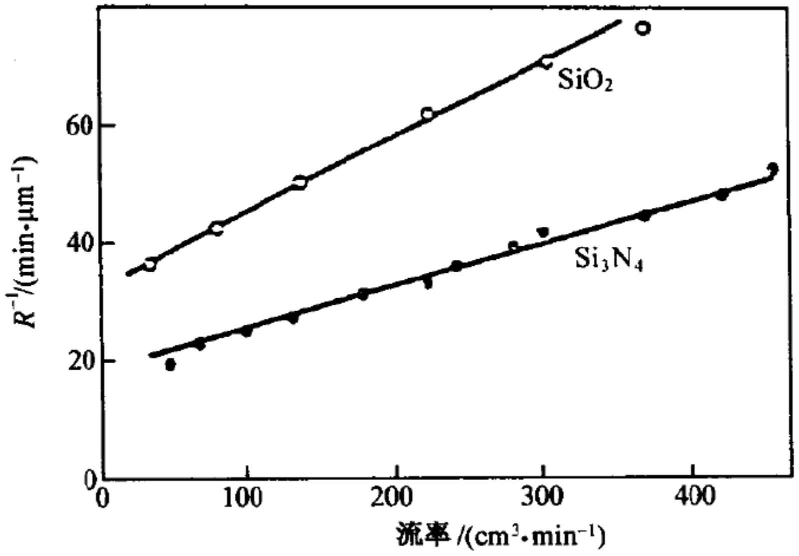
Use CF_4 or Cl_2 to etch silicide
it is isotropic to Poly-Si

Use Cl_2, HCl, $SiCl_4$ to etch Poly-Si
when Cl_2-Ar RIE is used, A=1



Effect of gase flow rate





- Generally, the effect gas flow rate on etching rate is small. But in the extreme case, it has an effect.

- When flow rate is very small, supplying of gas is limited; and when flow rate is very large, the output loss will have an effect, particularly, for the gas which has relative long reaction time.

Effect of gas pressure

- Reactive Ion Etching: 10 to 100 mTorr

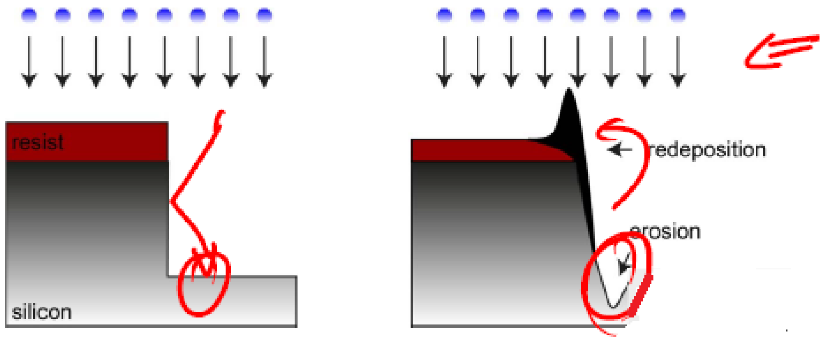
- HDP systems: 1 to 10 mTorr

Increasing the pressure causes more gas phase collisions to occur, decreasing the directionality of the etching.

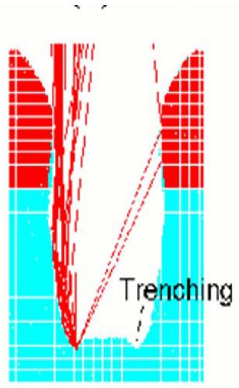
Increasing the pressure also increases the plasma density, up to a certain point

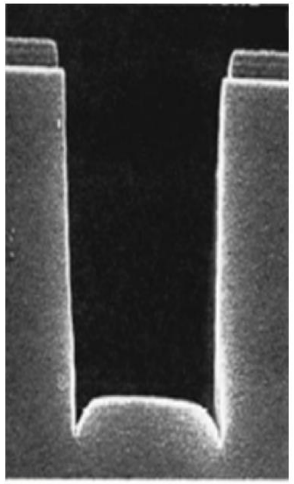
Above a certain pressure, the collisions between the gas molecules and electrons limit the energy of the electrons and thus limit the ionization rate

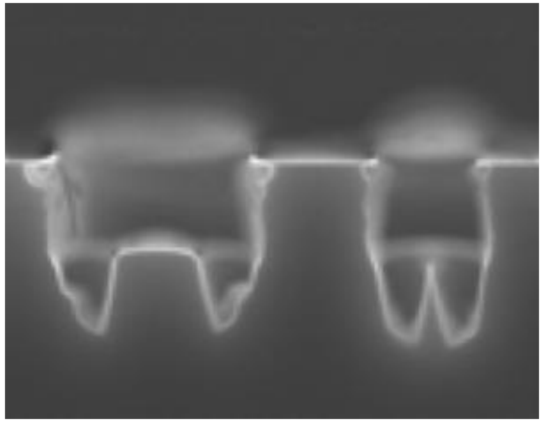
Deleterious RIE Effects: "Microtrenching"



- erosion at bottom surface due to reflection of ions at etch side walls (mechanical interaction
- if physical component dominant: redeposition of material at side wall and surface

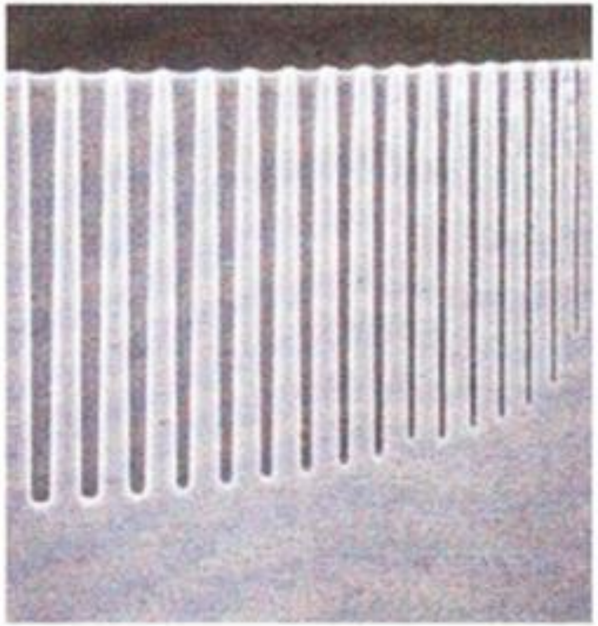


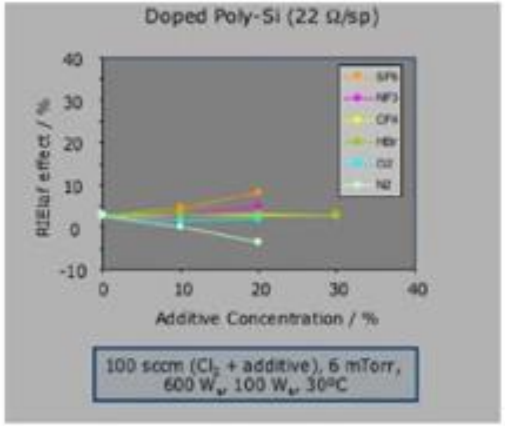




Pattern-Dependent Effects:

RIE Lag





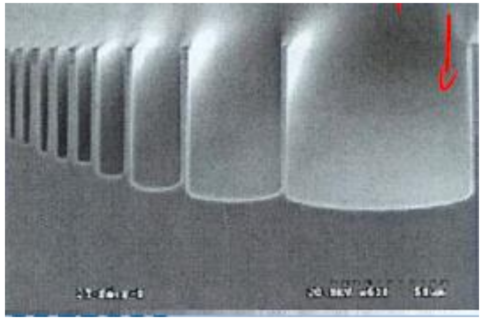
Pattern-Dependent Effects-microloading

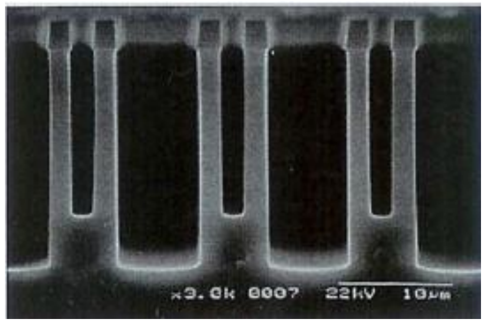
Microloading:

definition: micro loading describes the dependence of etch rate on structure with varying area density

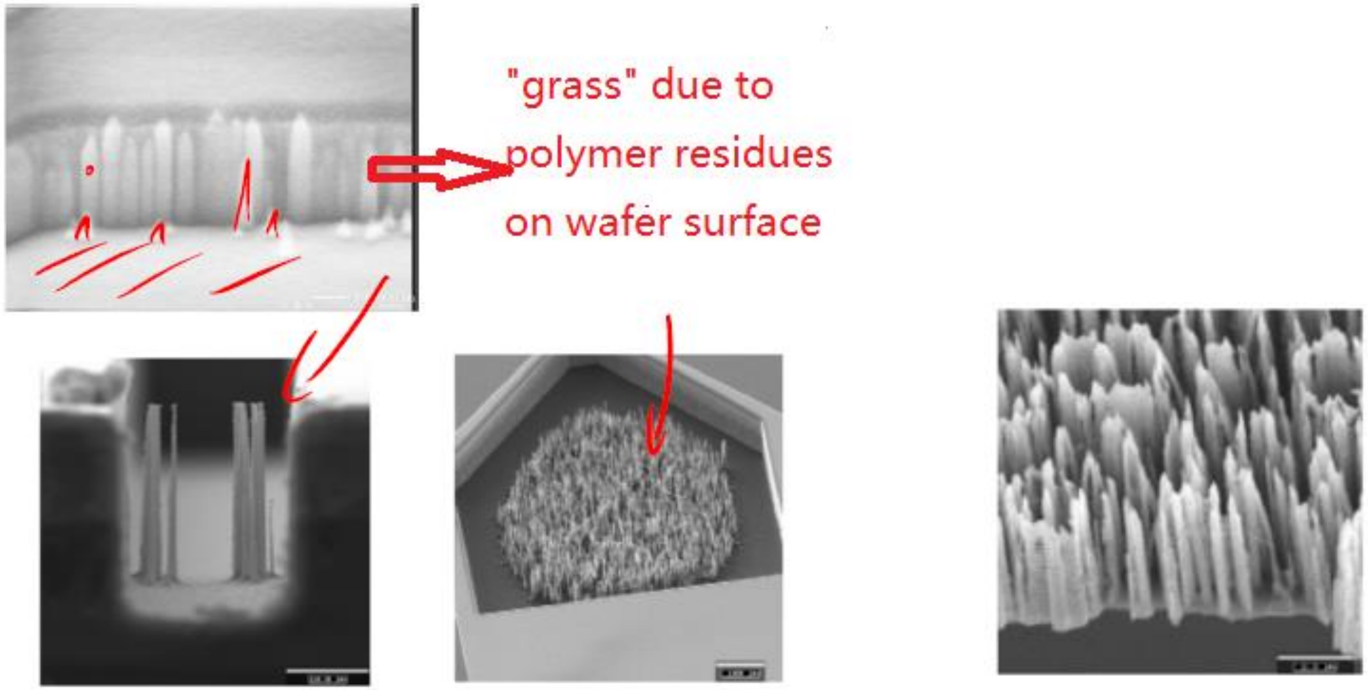
this means: lateral extended structures are etched faster than small structures

Reason: Etch species can attack surface more easily at larger structures; therefore higher etch rate





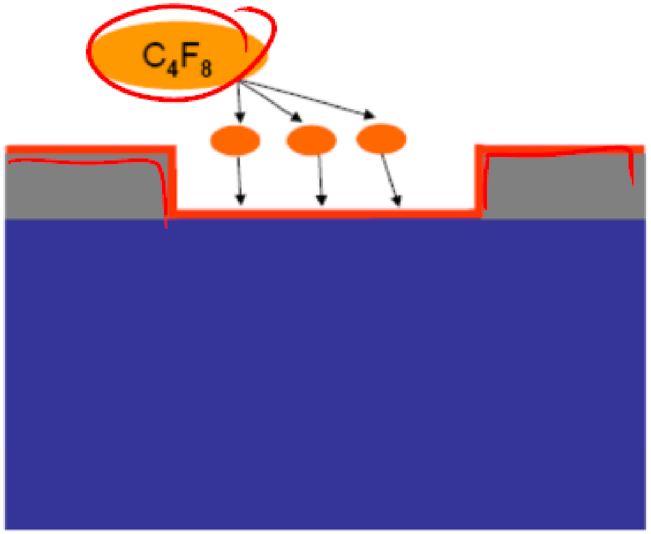
Dry etching effect-“grass”



Deep reactive etching-Bosch Process

With the rapid development of VLSI technology and MEMS technology, more and more devices require high AR microstructures. E.g., in MEMS Devices，deep silicon structures are the key Components in microsensors and microactuators.

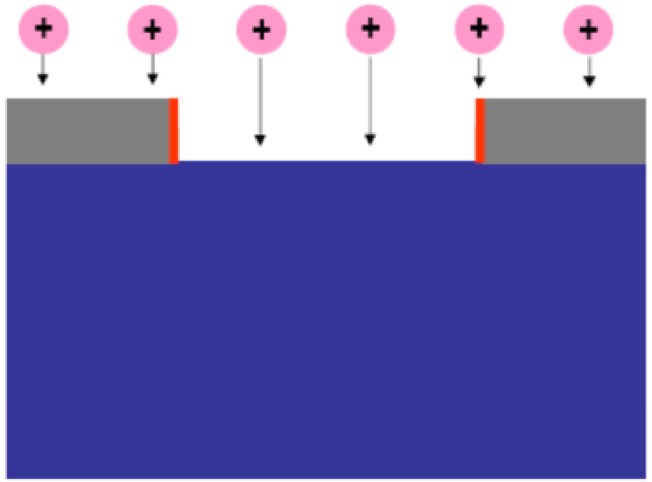
Deep reactive etching-Bosch Process



Deposition of thin polymer layer

- **$C4F8$, decomposed in plasma**
- **Formation of (CF2)n ploymer chains (order of few nm)**

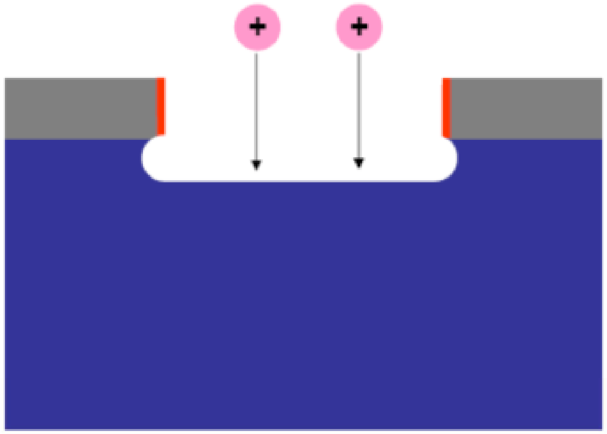
Deep reactive etching-Bosch Process



SF_6

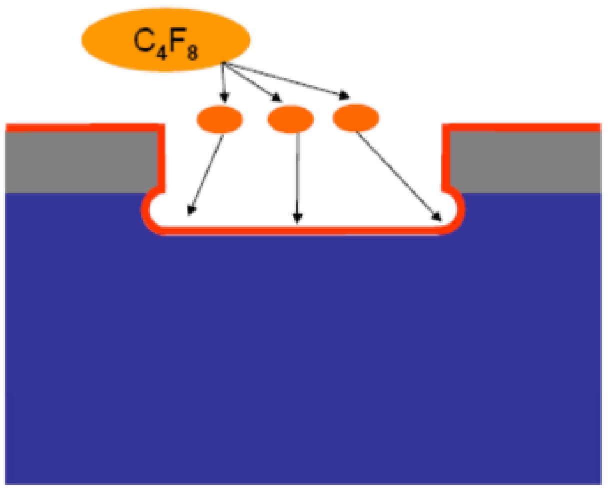
- Removal of polymer on surfaces
- Physical sputtering component only
- Very short etching time

Deep reactive etching-Bosch Process



- Isotropic etching with SF6, high selectivity, chemical component
- Ion assisted etching

Deep reactive etching-Bosch Process



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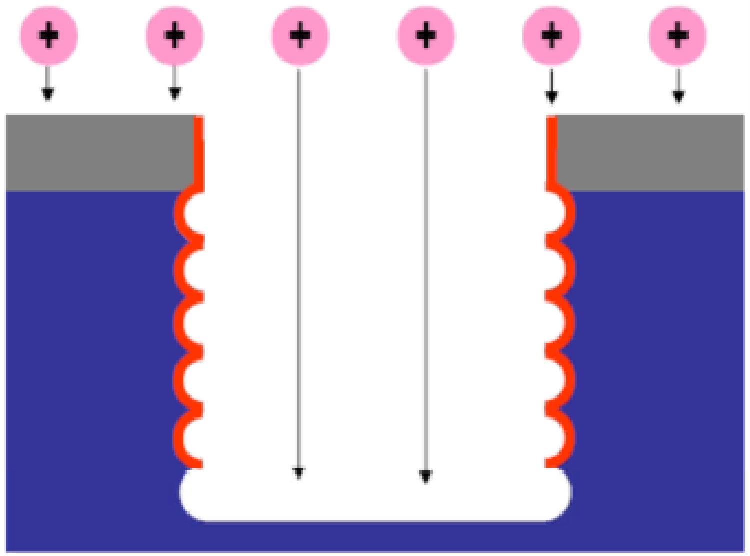
Deep reactive etching-Bosch Process

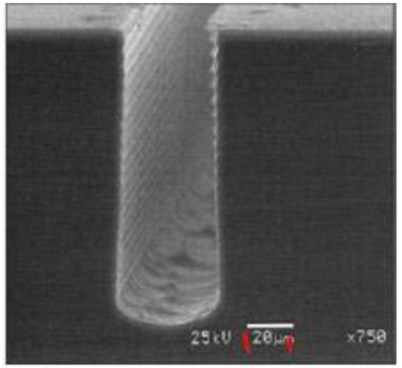


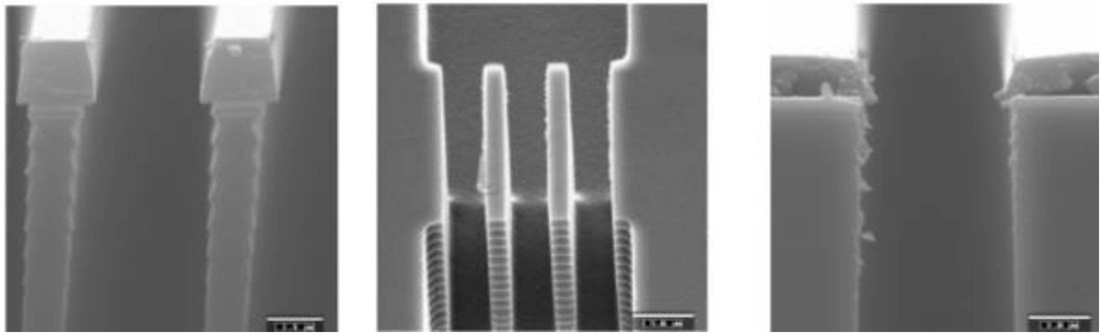
Repeat steps several times until desired etch depth

Deep reactive etching-Bosch Process

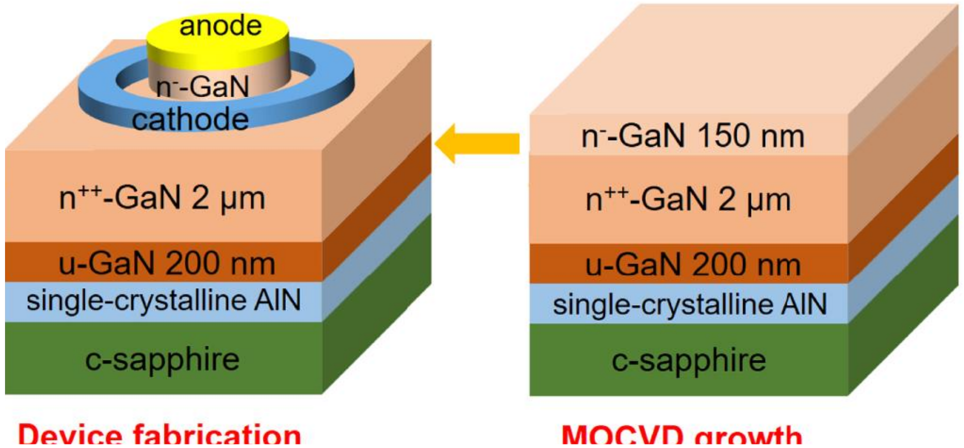
- Side-wall roughness due to isotropic etch step
- Can be diminished with larger number of etch steps

(increases process time)





Dry etching of GaN



离子物理轰击溅蚀
化学反应腐蚀

单面抛光蓝宝石衬底上生长的ＧａＮ

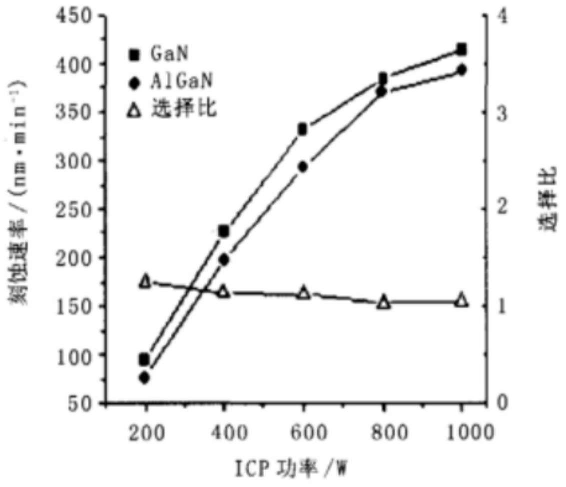


图 1　GaN 和 AlGaN 刻蚀速率、选择比随 ICP 功率的变化

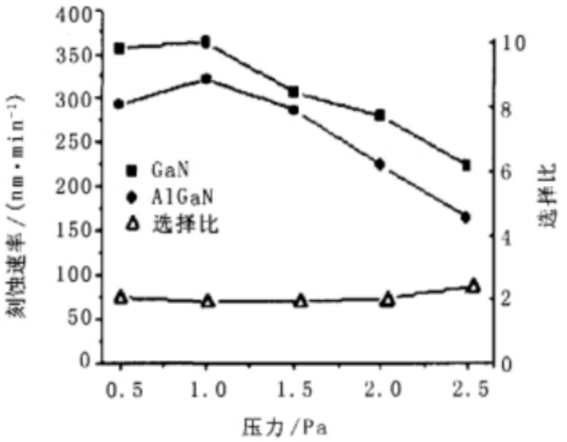


图 3　GaN 和 AlGaN 的刻蚀速率随反应室压力的变化

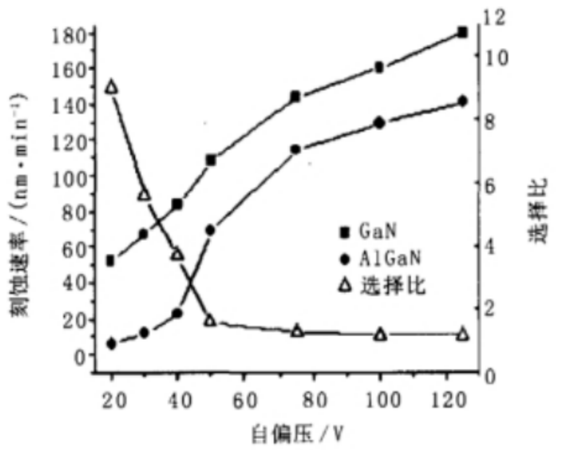


图 2　GaN 和 AlGaN 刻蚀速率、选择比随自偏压的变化

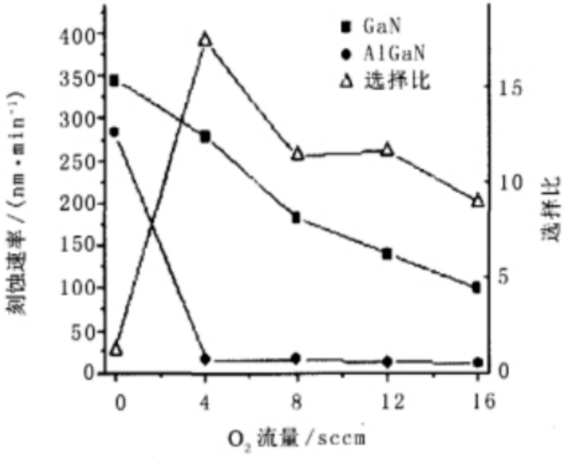


图 4　GaN 和 AlGaN 的刻蚀速率及选择比随 O_2 流量的变化

Outline

Summary

- **Basics of etching**

Anisotropy, selectivity, uniformity and requirements of etching

- **Wet etching**

Concept, characteristics, examples, drawbacks

- **Dry etching(mechanism)**

Concept, chemical etching, physical etching, RIE,

etching systems

- **Etching rate**

How some factors influence etching rate?

Summary

Wet vs. Dry Etching

	Wet	Dry
Method	Chemical solutions	Ion bombardment / Chemical reaction
Environment & Equipment	Atmosphere, Bath	Vacuum
Key Advantages	Low cost Easy to implement High selectivity	High lateral resolution
Key Disadvantages	Poor lateral resolution Chemical hazards Contamination	High cost Low etch rates Limited selectivity Substrate damage Toxic gases
Directionality	Isotropic (mostly)	Anisotropic (mostly)

homework

Example A 0.5-μm layer of silicon dioxide on a Si substrate needs to be etched down to the Si. Assume that the nominal oxide etch rate is r_{ox} (in μm min^{-1}). There is a ± 5% variation in the oxide thickness and a ± 5% variation in the oxide etch rate. (a) How much overetch is required (in % time) in order to ensure that all the oxide etched? (b) What selectivity of the oxide etch rate to the Si etch rate is required so that a maximum of 5.0 nm of Si is etched? (Assume the overetch calculated in part (a) is done.)